

Delegating Innovation: Sequential and Parallel Search Under Competition

Zhi Chen, Zhenyu Hu

NUS Business School, National University of Singapore, Singapore 119245.
zhi.chen@nus.edu.sg, bizhuz@nus.edu.sg

Yufei Zhu

Institute of Operations Research and Analytics, National University of Singapore, Singapore 117602.
yufei_zhu@u.nus.edu

Abstract: Concept generation plays a critical role in the success of new product development (NPD) projects by identifying products or services that address unmet customer needs. Given the inherent uncertainty and complexity of the innovation landscape, it is essential for firms to search broadly across the space of possible concepts to enhance the likelihood of discovering a winning idea. In practice, sequential and parallel search strategies represent two fundamental approaches to concept generation: sequential search develops and evaluates one concept at a time, stopping once a promising idea is found, while parallel search explores multiple concepts simultaneously before selecting the best. This paper considers a setting where a buying firm delegates the search process to multiple suppliers. In such a competitive, multi-agent setting, should the buyer prefer suppliers to adopt sequential or parallel search strategies? By embedding a classical search model into a contest framework, we show that the buyer prefers sequential search when either (i) the unit search cost is high or (ii) the maximum number of evaluation opportunities is large. In these cases, sequential search confers two advantages: an informational benefit that allows suppliers to adapt using performance feedback (a “quality effect”), and an incentive benefit that encourages suppliers to search more extensively (a “quantity effect”). When neither condition holds, parallel search is preferred. Solving the buyer’s optimal design problem further shows that the optimal supplier base always features restricted entry, limiting participation to as few as two suppliers. We also study an extension in which the firm incurs costs when evaluating submitted samples.

Key words: new product development, delegated search, contest

1 Introduction

Innovation is essential to a firm’s success, because it allows businesses to differentiate themselves and to establish a competitive edge. One of the key drivers that enables the development of innovative products and services is concept generation (Ulrich and Eppinger 2012, Kavadias and Hutchison-Krupat 2020). However, managing the concept generation process is challenging, as it is associated with significant complexity and uncertainty, making the search for a successful concept akin to locating a high peak within a rugged landscape (Sommer and Loch 2004, Terwiesch and Xu 2008). To navigate this challenging terrain, it is essential that firms thoroughly search across the space of possible product concepts, thereby increasing their chances of discovering a winning idea (Ulrich and Eppinger 2012). In practice, sequential and parallel search strategies represent two fundamental approaches to concept generation (Sommer and Loch 2004, Ulrich and Eppinger 2012) and NPD in general (Loch et al. 2001, Oraopoulos and Kavadias 2014, Shalpegin et al.

2018, Schlapp and Schumacher 2022). In a sequential search, firms search and evaluate one concept at a time and stop the search once a promising solution is identified (Weitzman 1979, Sommer and Loch 2004). In contrast, a parallel search involves setting a target number of concepts in advance and then exploring them simultaneously before identifying the best one (Dahan and Mendelson 2001, Sommer and Loch 2004).

However, firms oftentimes do not conduct concept generation internally but instead delegate it to external agents, such as suppliers or service providers (Terwiesch and Loch 2004, Rahmani and Ramachandran 2021). To fully leverage the diversity of ideas and approaches, firms (“buyer”) commonly engage multiple suppliers, each independently tasked with searching for novel concepts for the innovation challenge and bearing their own research and development (R&D) costs (Taylor 1995, Terwiesch and Xu 2008). In the end, the buyer is generally interested in acquiring the best solution, awarding the production contract to the supplier who proposes it. In such a multi-agent, competitive setting, a key question arises: Should the buyer (“she”) prefer each supplier (“he”) to conduct a sequential or parallel search? In this paper, we employ a contest approach to answer this question.

The food flavor industry helps exemplify our research setting and the question we study. Consumers have increasingly placed a high value on taste and sensory experiences, driving the need for companies to create innovative flavors that are in line with evolving consumer preferences and emerging trends (Spence 2015). This need has partly led to the food flavor industry with a global market size of 20.2 billion USD as of 2023 (Statista 2024). In addition to food and beverages, specialty flavors are also used in pharmaceuticals, nutritional supplements, and personal care products (International Flavors & Fragrances Inc. 2024). The development of specialty flavors typically requires a high degree of customization and technical expertise. A case study by Thomke and Nimgrade (2000) offers detailed insights into the flavor development process, which we summarize below.

When food companies (the “buyers”) such as Nestlé or Unilever seek to launch a new product, they typically begin by outlining preliminary inspirational specifications—such as the desired “flavor profile application” or how the final product should taste (p. 6). However, a key challenge lies in the subjective and imprecise language used to describe flavors, making the development process feel more like an art than a science (p. 1). To translate these abstract specifications into actual flavors, buyers often engage multiple flavor companies (“suppliers”) (p. 7). Each supplier embarks on a trial-and-error process, exploring different creative directions by developing different sample flavors (p. 17) to increase the likelihood of meeting the buyer’s expectations. Flavor development generally follows four key steps: extraction, analysis, formulation, and testing. The process starts with extracting and analyzing hundreds of compounds from source materials, then formulating the desired flavor using a few selected compounds. It concludes with application trials to evaluate how well the flavor aligns with the customer’s preferences when used in their final product (p. 17). In this process, suppliers bear the full cost of R&D, which can vary significantly depending on whether the project involves minor tweaks to existing flavors (low cost) or the development of entirely new flavor

families (high cost). Suppliers are only compensated through a production contract if their proposed flavor is ultimately selected and brought to market (p. 7).

The example above highlights two key features central to our study. First, the innovation task is characterized by ambiguity, open-endedness, and a subjective, more-art-than-science nature. This leads to an unstructured solution space that necessitates *trial-and-error search* during development (Thomke 2003b). Second, to leverage the diversity of proposed solutions, the buyer delegates the concept generation task to multiple suppliers who compete for an award in the form of a production contract, resembling a *contest* structure. These two features are also frequently observed in other creativity- and design-driven domains, such as automotive design (Langner and Seidel 2009), architecture (Terwiesch and Loch 2004) and marketing campaigns such as advertising (Swift 2014).

In addressing our research question on the buyer's preference between sequential and parallel search strategies, it is important to note that while the buyer may not directly dictate suppliers' search strategies, she can influence them. To see this, in many practical scenarios, only the buyer can evaluate the ultimate performance of a concept (Terwiesch and Loch 2004, Rahmani and Ramachandran 2021). In our aforementioned example, only the "applicationist" at Unilever, for instance, is able to tell whether the final product has the desired "green and leafy strawberry" taste by baking a strawberry shortcake in a pilot kitchen (Thomke and Nimgrade 2000). This unique evaluative role enables the buyer to actively engage in the suppliers' development process by providing performance feedback on how well a proposed solution aligns with her preferences (Terwiesch and Loch 2004, Rahmani and Ramachandran 2021). In this way, the buyer guides suppliers towards a *sequential search* process, where each supplier can utilize the performance feedback to decide whether to continue developing new solutions or to stop.

Alternatively, the buyer may take a more hands-off approach by not holding interim meetings with suppliers, instead conducting a single final evaluation of all proposed solutions at once, an approach sometimes recommended in creative collaborations such as advertising development (Koslow et al. 2006). This setup resembles a *parallel search* environment. As suppliers receive no feedback during development, they can ex-ante determine the number of solutions to generate and work on each solution in a parallel fashion until the final evaluation.

The trade-off between sequential and parallel searches is well-understood in a single-agent context (Loch et al. 2001, Sommer and Loch 2004): Parallel search tends to be faster, as it completes the development process in a single stage rather than spreading it across multiple sequential iterations (Loch et al. 2001). In contrast, sequential search enables learning from previous outcomes (Sommer and Loch 2004) and can be more cost-effective if successes emerge early in the search process (Loch et al. 2001). However, in our setting, the incentive dimension—how competition affects suppliers' search efforts and ultimately the buyer's profit under different search strategies—becomes a central consideration, one that is largely absent in single-agent analyses.

Motivated by the previous discussions, our paper integrates the classical search model (Lippman and McCall 1976), commonly used to model innovation processes (e.g., Taylor 1995, Terwiesch and Loch 2004, Rahmani and Ramachandran 2021), within a contest framework. Here, each supplier incurs a unit search cost to generate a solution, whose performance value is modeled as a random draw from a known probability distribution. Under a sequential search, the buyer periodically evaluates the submitted solution and communicates its value to the focal supplier. Based on this feedback, the supplier decides whether to continue searching for a new alternative or to stop. If the supplier stops, the best solution he has developed thus far is treated as his final submission. In contrast, under a parallel search, suppliers *ex ante* determine the number of solutions to pursue and develop them simultaneously within a single round, submitting all solutions at once for the buyer’s evaluation.

1.1 Summary of Main Findings

We can summarize our main findings as follows. Whether sequential search or parallel search is more profitable depends crucially on two tangible factors closely tied to the characteristics of the R&D process: (i) unit search cost, and (ii) the maximum number of evaluation opportunities. The detailed findings are summarized in Table 1 below.

Unit search cost		Low	High
Number of evaluation opportunities	Few	Parallel Search	Sequential Search
	Many	Sequential Search	Sequential Search

Table 1 Buyer’s preferred search strategies

Specifically, sequential search is more profitable for the buyer, either when there are many evaluation opportunities or when unit search costs are high. We show that, in a competitive setting, sequential search offers two key advantages: an informational advantage that allows suppliers to adapt their search based on the performance feedback (“quality effect”), and an incentive advantage that motivates suppliers to conduct more searches (“quantity effect”). While the quality effect is well-established in the single-agent context, the quantity effect is novel to the literature, enriching our understanding on how competition affects search incentives under each search strategy. On the other hand, when there are few evaluation opportunities and search costs are low, sequential search is inferior to parallel search. Here, due to the low search cost, suppliers tend to fully utilize all available evaluation opportunities by conducting the maximum possible number of searches under parallel search. However, the limited number of evaluation opportunities diminishes the two key benefits of sequential search, the quantity and the quality effects, making it inferior to parallel search.

The previous findings apply to settings in which both the number of evaluation opportunities and the number of suppliers are fixed. For example, in many industries, the number of eligible suppliers is limited

and buyers cannot freely invite new suppliers outside their approved vendor list. However, what if the buyer wants to reconfigure her supplier base? To this end, we address the buyer's optimal design problem, where she can jointly choose the number of suppliers to engage, number of evaluation opportunities to grant, and which search strategy for suppliers to employ. We find that the buyer should always steer suppliers toward sequential search and commit all available evaluation resources. Moreover, the optimal supplier base is always restrictive entry (as opposed to open entry), which is consistent with industrial procurement practices.

In our main model, we assume that the buyer incurs no evaluation cost. In an extension, we introduce a positive unit evaluation cost and revisit the buyer's preference between sequential and parallel search. We find that when the evaluation cost is low relative to the unit search cost, the buyer's preference established in the main model continues to hold. When the evaluation cost is sufficiently high, however, the buyer's preference reverses. We also study the buyer's optimal design problem in the presence of evaluation costs and show that the optimal design depends critically on the magnitudes of both the unit search cost and the unit evaluation cost.

1.2 Managerial Implications

Our main findings have direct managerial implications. Regarding the two key factors outlined in Table 1, search costs are often associated with the complexity of the innovation. In the example of food flavor development, projects involving minor modifications to existing flavors incur relatively low search costs (a few thousand dollars), whereas projects requiring the creation of entirely new flavor families can be significantly more expensive, with costs reaching up to \$300,000 (Thomke and Nimgrade 2000, p. 7). More broadly, innovation projects that require only virtual and simple prototypes (e.g., logo design) generally involve lower search costs than those requiring physical and complex prototypes (e.g., an automotive system).

The number of evaluation opportunities refers to the maximum number of solutions that the buyer can evaluate for each supplier, which depends on, for example, the buyer's available personnel and equipments. To illustrate, in the food-flavor example, flavors developed by suppliers undergo "application trials" conducted by "applicationists" and, in some cases, may even require the final product to be baked in a pilot kitchen. Hence, in this case, the number of evaluation opportunities granted to suppliers depends critically on the total number of applicationists and pilot kitchens available to the buyer.

Our results show that for projects with low search costs (e.g., requiring development of only virtual prototypes or modification of existing flavors) and few evaluation opportunities (e.g., when the buyer has limited personnel or pilot-kitchen capacity), the buyer prefers that suppliers conduct parallel search. When either search costs are high or when there are many evaluation opportunities, the buyer prefers that suppliers conduct sequential search. Overall, our managerial discussions above help managers overseeing delegation of innovation projects to conceptualize and map their project characteristics into the framework in Table 1, ultimately enabling them to guide suppliers towards the appropriate search processes. We also solve the

buyer's optimal design problem and extend the model to allow for positive evaluation costs, such as the monetary costs incurred when evaluating prototypes, thereby offering further insights into how the search process can be structured to improve innovation outcomes.

2 Related Literature

Concept generation and testing are two core activities in many NPD and innovation projects (Kavadias and Hutchison-Krupat 2020). Due to the uncertain nature of NPD, the academic literature has conceptualized both activities as a search process. The study of *concept testing* began with the seminal work of Weitzman (1979). The setup in Weitzman (1979) examines a sequential search problem in which a decision-maker must choose the best option from a set of alternatives, each characterized by an ex ante uncertain and heterogeneous payoff whose true value can be revealed at a known search cost. The goal is to determine the order of testing as well as when to stop. While Weitzman (1979) and subsequent work (e.g., Erat and Kavadias 2008, Bhaskaran and Erat 2025) study sequential testing, Dahan and Mendelson (2001) advocates for parallel testing, where alternatives are evaluated simultaneously. Building on earlier work, Loch et al. (2001) highlights a key trade-off between the parallel and sequential testing. While the former offers speed, it limits the utilization of information generated during testing, potentially leading to a greater number of tests and higher costs. In contrast, the latter allows for adaptive strategies and may reduce testing costs, though it often requires more time. Shalpegin et al. (2018) and Schlapp and Schumacher (2022) identify additional factors that influence the relative performance of the two testing strategies when testing is delegated to external agents. For instance, Shalpegin et al. (2018) show that the supplier's private information about the production cost makes parallel testing less attractive.

Notably, in contrast to the prior literature on concept testing, our work falls within the category of *concept generation*. Whereas concept testing assumes a fixed set of ex ante heterogeneous alternatives and focuses on determining the optimal sequence in which to test them, concept generation typically starts with no existing options and emphasizes the creation of new concepts, each at a cost drawn from a distribution. The central decision in concept generation is therefore not a sequencing problem, but a stopping problem, that is, when to stop generating new concepts and select the best alternative discovered so far.

To capture the inherently uncertain and trial-and-error nature of concept generation process (Terwiesch and Loch 2004), researchers have drawn on the well-known search models (Lippman and McCall 1976).¹ Directly related to our study are papers that examine delegated search. Rahmani and Ramachandran (2021) study whether a client should adopt a committed or flexible stopping policy when an agent conducts sequential search on the client's behalf, where the agent is compensated with a fixed payment upon discovering at

¹ Search models have been the workhorse in a wide range of applications such as job search (Zorc and Tsetlin 2020, Hu and Tang 2021, Huh et al. 2025), matching (Lin and Tang 2025), consumer search (Hu and Xiao 2024), and project management (McCardle et al. 2018).

least one feasible solution. In a related sequential search setting, Zorc et al. (2025) analyze when a principal should delegate the search process to an agent and how to design an optimal contract, with compensation contingent on the value of the solutions discovered. While details differ, both papers share a key feature in common: the value of the solution is *verifiable*. As a result, the principal can either specify a feasibility threshold (Rahmani and Ramachandran 2021) or make compensation contingent on the realized solution value (Zorc et al. 2025), which provide search incentives for the agent.

However, in the settings we consider (such as flavor development or creativity- and design-driven domains), solution quality is often *non-verifiable*, rendering the incentive instruments used in the prior literature, such as feasibility thresholds or contingent contracts, inapplicable. Consider flavor development as an illustrative example. It is virtually impossible for the buyer to specify *ex ante* what constitutes a feasible outcome, nor can contract payments be conditioned on the realized “value” of a developed flavor. In such settings characterized by the absence of verifiable performance metrics, contests have been identified as an effective mechanism for providing innovation incentives both in theory and in practice (Deng and Elmaghraby 2005, Chen et al. 2022b, Fan et al. 2024). At the outset of the contest, the buyer commits a prize (e.g., a supply contract) to be awarded to the supplier submitting the best solution. Since the prize must be allocated to one of the suppliers regardless, the buyer has no incentive to deviate from selecting the solution she prefers the most. Meanwhile, competition for the contest prize incentivizes effort from the supplier base.

Inspired by the benefits of contests discussed above, the extant literature has examined the use of contests to motivate suppliers’ search incentives by embedding the search process within a contest framework. The two papers most closely related to ours are Taylor (1995) and Terwiesch and Xu (2008), which analyze suppliers’ incentives under sequential and parallel search, respectively. However, no prior work has compared the relative performance of the two search strategies from the buyer’s perspective. Addressing this gap is the focus and main contribution of our study.

Within the broader contest literature, a closely related stream of work studies how interim performance feedback can be used to manage contests dynamically (Bimpikis et al. 2019, Mihm and Schlapp 2019, Ely et al. 2022).² In our model, the buyer’s interim performance feedback induces suppliers to engage in sequential search. Thus, our model of sequential search under competition can also be interpreted as a contest in which the buyer provides *private feedback*, that is, performance evaluations are shared only with the individual supplier and remain undisclosed to competitors. Similarly, our parallel search model corresponds to a contest setting with *no feedback*. Consequently, our central research question, whether the buyer should

² Beyond dynamic contests, there is also a large body of work examining how various design levers affect contest outcomes in static settings (see Ales et al. 2019 for an excellent review). These include number of suppliers allowed in contests (Körpeoğlu and Cho 2017, Ales et al. 2021), the role of participation (Stouras et al. 2022), award structure (Moldovanu and Sela 2001, Ales et al. 2017), contest duration (Körpeoğlu et al. 2021), problem specifications (Erat and Krishnan 2012, Jiang et al. 2021), whether to allow team submissions (Candoğan et al. 2025), contestant screening (Khorasani et al. 2023), management of multiple parallel contests (Körpeoğlu et al. 2022, Stouras et al. 2025) and the comparison between separate contests and a joint one (Hu and Wang 2021).

prefer suppliers to conduct sequential versus parallel search, can be reframed as a comparison of the buyer’s profits in contests with private versus no feedback. However, the majority of research studying the effect of feedback in contests has analyzed the effects of *public feedback* (see, e.g., Bolton and Harris 1999, Aoyagi 2010, Ederer 2010, Wooten and Ulrich 2017, Jiang et al. 2022), where all participants’ interim performances are publicly disclosed, as exemplified by leaderboards in crowdsourcing platforms like Kaggle.³ In industrial procurement, which is our focal setting, we are not aware of any setting that employs public feedback in contests, perhaps due to the proprietary nature of industrial R&D.

A related work is Mihm and Schlapp (2019), which compares the effects of commonly used feedback policies (including private and no feedback) using an “effort plus noise” model. In contrast, we specifically consider the supplier’s sequential search process, a common feature in industrial R&D. By viewing the number of searches as an effort decision, Ales et al. (2017, Lemma 1, p. 695) shows that the uncertainty in the performance value from each search can be translated to an additive noise when the performance value is drawn from a Gumbel distribution. However, such an equivalence is not true under a general distribution and one cannot simply use the “effort plus noise” model to capture the search process. Moreover, even if we restrict the uncertainty in the performance value in our model to follow a Gumbel distribution, our sequential search model is conceptually different from that of Mihm and Schlapp (2019). To see this, motivated by the repetitive search process in which each of the suppliers develops multiple solutions over time (e.g., as seen in the food flavor development), a supplier’s final performance in our model is determined by the *maximum* performance attained across rounds. This implies that the performances in later rounds *do not build on* those in earlier ones, which explains why suppliers’ search incentives are governed by a threshold policy (Taylor 1995). In contrast, this relationship is assumed *additive* in Mihm and Schlapp (2019), where performance accumulates over time. This implies a *build-on* effect across rounds, which explains why a supplier’s efforts in later rounds depend on his specific realized performances in previous rounds (Mihm and Schlapp 2019). As a result of this conceptual difference, the two papers offer distinct insights.

2.1 Contributions to the Literature

In sum, our study bridges two rich streams of research in NPD, the literature that views innovation processes through the lens of search and the innovation contest literature. While the search literature emphasizes informational efficiency and adaptation, and the contest literature focuses on incentive provision, we demonstrate that the interaction between the two offer novel insights that neither literature predicts on its own. Specifically, with respect to the search literature, prior studies highlight time efficiency as a key advantage of parallel search. Accordingly, in centralized settings, sequential search is known to be optimal when time costs are negligible (Loch et al. 2001). In contrast, we show that in delegated-search settings where incentives matter,

³ In addition to interim feedback, contest organizers may also provide public feedback on the performance of final submissions, even while other suppliers are still in progress (Chen et al. 2022a).

even in the absence of time costs, the buyer sometimes prefers parallel over sequential search because it provides stronger incentive effects that outweigh the informational advantage of sequential search. This highlights how incentive considerations arising from contests can alter prior insights in the search literature.

With respect to the contest literature, in an effort-plus-noise model, Mihm and Schlapp (2019) show that performance uncertainty is the key driver that determines whether feedback helps improve the contest outcomes. In contrast, the key contingencies in our paper are search costs and the maximum number of evaluation opportunities or rounds. In addition, our general multi-supplier, multi-round framework provides new insights into how the two key parameters: maximum number of evaluation opportunities or rounds and the size of the supplier base, influence the choice of feedback policies, as well as how the buyer can utilize these factors to enhance contest outcomes. Since Mihm and Schlapp (2019) is based on a two-round (i.e., two evaluation opportunities), two-supplier setup, the effects of these two parameters are absent from their analysis. Therefore, our multi-supplier, multi-round framework addresses this gap. This underscores the importance of adding search-based features to the contest model.

3 Model Setup

We consider a setting in which a buyer delegates a customized innovation task to $n \geq 2$ suppliers, each of whom is granted a maximum of $R \geq 2$ evaluation opportunities. The parameter R could be interpreted as the maximum number of solutions the buyer is able to evaluate from each supplier. At the outset of the development process, a fixed award worth $A > 0$, typically in the form of a production contract, is announced and will be offered to the supplier who delivers the best solution. Given that at most R solutions will be evaluated for each supplier, we label the value of the solution developed by supplier i that will be evaluated in the r -th evaluation opportunity as $X_{i,r}$, which is assumed to be drawn independently and identically from a commonly known distribution $F(\cdot)$ with finite mean and strictly positive density $f(\cdot)$ on the interval $[0, 1]$.

3.1 Parallel Search

When suppliers adopt parallel search strategies, each supplier decides at the outset whether to participate in the competition and, if so, a fixed $m_i \in \{1, 2, \dots, R\}$ number of solutions to develop simultaneously. The developed solutions are then submitted to the buyer for evaluation. Since the buyer is only interested in the best solution, the resulting performance value of supplier i is then given by $X_{i,(m_i)} = \max_{1 \leq r \leq m_i} X_{i,r}$. In this paper, the highest order statistic within m samples is denoted by the standard notation $X_{(m)}$.

Given that the opponent suppliers develop $m_{-i} = (m_j)_{j \neq i}$ solutions, supplier i 's expected payoff is given by $\pi_i(m_i | m_{-i}) = A \cdot \mathbb{P}(X_{i,(m_i)} > X_{j,(m_j)}, j \neq i) - cm_i$. We will abbreviate $\pi_i(m_i | m_{-i})$ as $\pi_i(m_i | m)$ if $m_j = m$ for all $j \neq i$. Here, supplier i will only be the winner and obtain the award A if his final performance value is better

than that of all opponents.⁴ However, supplier i always pays the search costs cm_i , regardless of whether he wins. A strategy profile $(m_1^*, \dots, m_n^*) = (m^*, \dots, m^*)$ is a symmetric pure-strategy equilibrium of the game if

$$m^* \in \operatorname{argmax}_{m_i \in \{1, 2, \dots, R\}} \pi_i(m_i | m^*).$$

The buyer's profit when suppliers adopt parallel search strategies is then given by:

$$\Pi_P = \mathbb{E} \left[\max_{i=1, \dots, n} X_{i, (m^*)} \right] - A. \quad (1)$$

In our main model, we assume that the buyer incurs no cost when evaluating suppliers' solutions. In Section 6, we extend the model to consider a positive unit evaluation cost incurred by the buyer.

3.2 Sequential Search

In comparison to the above setting where all suppliers search simultaneously first and then the buyer evaluates all submitted solutions, the buyer can actively involve in the suppliers' development process by dividing the R evaluation opportunities into R evaluation rounds. In each round, each supplier develops a solution and then submits it to the buyer, who then evaluates it and provides performance feedback about the value of the submitted solution to the supplier. This allows suppliers to conduct their search sequentially, utilizing the feedback information from each round to decide whether to continue or to stop.

More formally, once supplier i searches in evaluation round r , the realization of $X_{i,r}$ is privately communicated to him. In other words, at the beginning of each evaluation round r , the only state information supplier i has is his own best performance value so far, which we denote as $\hat{x}_{i,r}$, and each supplier is essentially deciding whether to continue to the next round or to stop searching based on his most up-to-date best performance value. Given any stopping times τ_j for suppliers $j \neq i$, we define $\Phi_i(\cdot)$ as the cumulative distribution function of the best performance value submitted by all of supplier i 's opponents at the end of the entire search process. That is, $\Phi_i(x) = \mathbb{P}(\max_{j \neq i} X_{j, (\tau_j)} \leq x)$. Additionally, we let $V_{i,r}(\hat{x}_{i,r} | \tau_{-i})$ represent supplier i 's value-to-go function at the beginning of the evaluation round r when his best performance value is $\hat{x}_{i,r}$. We then have the following Bellman equation that determines supplier i 's optimal stopping time τ_i :

$$V_{i,r}(\hat{x}_{i,r} | \tau_{-i}) = \max \{ A \cdot \Phi_i(\hat{x}_{i,r}), -c + \mathbb{E}[V_{i,r+1}(\max\{\hat{x}_{i,r}, X_{i,r}\} | \tau_{-i})] \}, \quad r = 1, \dots, R, \quad (2)$$

with the terminal condition $V_{i,R+1}(\hat{x} | \tau_{-i}) = A \cdot \Phi_i(\hat{x})$. A strategy profile $\tau^* = (\tau_1^*, \dots, \tau_n^*)$ is a symmetric equilibrium if $\tau_i^*, i = 1, \dots, n$ are independent and identically distributed, and for every i , τ_i^* is a solution to (2) given τ_{-i}^* .

⁴ We note that in cases where there are multiple winners, a common assumption is that the award is split equally among them. We do not explicitly write out such cases here since they occur with probability zero in the equilibrium.

The buyer's profit when suppliers adopt sequential search strategies is then given by:

$$\Pi_S = \mathbb{E} \left[\max_{i=1, \dots, n} X_{i, (\tau_i^*)} \right] - A. \quad (3)$$

Compared to the parallel search model introduced in Section 3.1, note that the total number of searches m^* in (1) is replaced by the stopping time τ_i^* , which is a random variable adapted to the realization of the process $X_{i,r}$, $1 \leq r \leq R$.

3.3 Assumptions and Discussion

The base model described above relies on several assumptions, which we examine below.

Sequential Search with No Learning Between Rounds. In our sequential search model presented in Section 3.2, we assume that the buyer discloses only the performance of the submitted solutions, without providing guidance on how suppliers can improve in future iterations, an assumption also made in Terwiesch and Loch (2004) and Rahmani and Ramachandran (2021). For example, as Terwiesch and Loch (2004, p. 148) noted “[t]he observation of the n th prototype does not provide any guidance as to how to choose the specifications for the $(n + 1)$ st prototype.” There are mainly two reasons for this. First, our focal setting concept generation phase is commonly linked to an unstructured solution space. Therefore, it is difficult for buyer to provide the exact guidance to supplier on how to improve. In the example of food flavor development, buyers often find it difficult to “articulate with words what they want in a flavor” (Thomke 2003a, p. 10). Second, even if in certain cases, “[t]he [buyer] may know precisely what that means, but flavorists can find the request difficult to interpret.” (Thomke 2003a, p. 4), making the buyer's guidance moot. In addition, the buyer may refrain from providing precise improvement directions because it may hinder supplier creativity (Koslow et al. 2006) or sometimes an improvement in one dimension can result in unintended negative consequences in other dimensions due to complex interactions (Sommer and Loch 2004).

Time Cost. In our model, suppliers complete development in a single round under parallel search, whereas development typically extends over multiple rounds under sequential search. Hence, sequential search leads to a slower time-to-market, entailing higher time costs from the buyer's perspective. However, in our model, we assume zero time costs. We make this assumption because the impact of time costs is well-understood in the literature—it tends to makes sequential search less attractive (Loch et al. 2001). By abstracting away from time costs in our base model, we can focus on how competition influences suppliers' search incentives under each search strategy. Unlike the single-agent setting in Loch et al. (2001), we will show that even with zero time cost, there still exists a trade-off between parallel search and sequential search in a competitive context.

No Early Termination in Sequential Search by the Buyer. Since time costs and evaluation costs are assumed to be zero in our main model, the buyer has no incentive to terminate the development process early under sequential search; instead, development continues until suppliers choose to stop. However, when time

costs or evaluation costs are non-negligible, the buyer must trade off between the potential improvement in solution quality from an additional round and the incremental time cost incurred, and therefore, may choose to terminate the search earlier before suppliers do.

Truthful Private Feedback in Sequential Search. In our sequential search model presented in Section 3.2, we assume that the buyer discloses *truthfully* the performance of the submitted solution, a common assumption in the information design literature (e.g., Mihm and Schlapp 2019, Khorasani et al. 2024, Xiao et al. 2024). In addition, the performance information is only *privately* communicated to the focal supplier and remains hidden from the competitors. This is aligned with practice (Liker et al. 1996, Thomke and Nimgrade 2000, Langner and Seidel 2009), due to the proprietary nature of R&D.

Finally, throughout the paper, we invoke the following technical assumption.

ASSUMPTION 1. *We assume that the unit search cost is smaller than the expected award received by each supplier, that is, $c < A/n$.*

This assumption is commonly invoked in the prior literature (Taylor 1995, Chen et al. 2022a). It ensures that all suppliers in our model participate in the search process and search at least once. This is in a similar spirit to assumptions made in other competitive settings that ensure agent participation (e.g., Ales et al. 2021, Chen et al. 2025, Körpeoğlu et al. 2022).

4 Equilibrium Analysis

Before comparing the buyer's profits when suppliers adopt the two distinct search strategies in Section 5, we first characterize the equilibrium number of searches for suppliers under each strategy. We begin with the case where suppliers search in parallel in Section 4.1, followed by the case where they search sequentially in Section 4.2.

4.1 Parallel Search

As mentioned previously, with parallel search strategies, suppliers decide at the outset how many searches to conduct, carrying them out simultaneously. The number of searches is capped by R , the maximum number of evaluation opportunities available to each supplier. In the following proposition, we fully characterize the symmetric equilibrium and provide conditions for when the equilibrium is in a pure or mixed strategy. Following the established notation, we use $\lfloor x \rfloor$ and $\lceil x \rceil$ to denote the floor and ceiling of x , respectively.

PROPOSITION 1. *Consider a setting with n suppliers, where each supplier adopts a parallel search strategy and has at most R evaluation opportunities. Then, a symmetric Nash equilibrium always exists. More specifically, with $m_0 = A(n-1)/(cn^2)$, we have:*

(i) *If $R \geq \lceil m_0 \rceil$, when the interval $[m_0 - 1/n, m_0 + 1/n]$ contains an integer s , $m^* = s$ is a symmetric pure-strategy Nash equilibrium. Moreover, if s is the only integer in that interval, then m^* is the unique symmetric equilibrium.*

(ii) If $R \geq \lceil m_0 \rceil$, when the interval $[m_0 - 1/n, m_0 + 1/n]$ does not contain any integer, the unique symmetric Nash equilibrium is in mixed strategy, given by:

$$m^* = \begin{cases} \lfloor m_0 \rfloor, & \text{with probability } p^* \\ \lceil m_0 \rceil, & \text{with probability } 1 - p^*, \end{cases}$$

in which $p^* \in (0, 1)$ is the unique p that solves:

$$A \cdot \sum_{i=0}^{n-1} \binom{n-1}{i} p^i (1-p)^{n-i-1} \cdot \left[\frac{\lceil m_0 \rceil}{n(\lceil m_0 \rceil) - i} - \frac{\lfloor m_0 \rfloor}{n(\lfloor m_0 \rfloor - i - 1)} \right] - c = 0.$$

(iii) If $R < \lceil m_0 \rceil$, the unique symmetric Nash equilibrium is in pure strategy, with $m^* = R$.

Propositions 1(i) and (ii) indicate that when each supplier is granted with many evaluation opportunities (i.e., $R \geq \lceil m_0 \rceil$), the equilibrium number of searches m^* depends on whether the interval $[m_0 - 1/n, m_0 + 1/n]$ contains an integer. Notably, the width of the interval $[m_0 - 1/n, m_0 + 1/n]$ is $2/n$. This width equals 1 when $n = 2$ and is strictly less than 1 when $n > 2$, meaning the interval can contain at most two integers. The case where two integers are contained is rare, occurring only when $n = 2$ and the lower bound of the interval $(A/(4c) - 1/2)$ is an integer. Proposition 1(i) shows that, except for these corner cases, if $[m_0 - 1/n, m_0 + 1/n]$ contains an integer s , that integer is the unique symmetric equilibrium. Proposition 1(ii) shows that when $[m_0 - 1/n, m_0 + 1/n]$ does not contain an integer, each supplier mixes between two adjacent integers, $\lfloor m_0 \rfloor$ and $\lceil m_0 \rceil$, in the unique symmetric equilibrium. In contrast, when each supplier has few evaluation opportunities (i.e., $R < \lceil m_0 \rceil$), Proposition 1(iii) demonstrates that $m^* = R$ is the unique symmetric equilibrium.

To understand the intuition behind the equilibrium, it is helpful to highlight two key drivers. First, the equilibrium number of searches, m^* , is determined by the trade-off between the marginal cost c and the marginal return from an additional search (the latter being a function of the award A and the number of suppliers n). Second, m^* may be constrained by R , the maximum number of evaluation opportunities available to each supplier. When R is large, as in Propositions 1(i) and (ii), m^* is driven solely by the economic trade-off and remains independent of R . However, when R is small, as in Proposition 1(iii), m^* is limited by R , even if suppliers are incentivized to conduct more searches.

Another notable feature of Proposition 1 is the interval structure $[m_0 - 1/n, m_0 + 1/n]$. If this interval contains an integer, that integer represents the unique symmetric equilibrium, as suppliers have no incentive to deviate due to the concavity of their expected payoff. However, when the interval does not contain an integer, suppliers find it profitable to deviate towards integers near the interval, making a pure-strategy equilibrium impossible. As Proposition 1(ii) shows, in such cases, suppliers adopt a mixed strategy, alternating between two adjacent integers (rather than any two integers), again driven by the concavity of their payoff function.

As mentioned, our parallel search model has been analyzed in Terwiesch and Xu (2008). Theorem 1B(iii) of Terwiesch and Xu (2008) characterizes the equilibrium number of searches conducted by each supplier as $m_0 = A(n-1)/(cn^2)$, which may take fractional values. Proposition 1 extends this equilibrium result from

Terwiesch and Xu (2008) in two important ways. Conceptually, our analysis explicitly accounts for the reality that a supplier can only conduct an integer number of searches and that the buyer may be able to evaluate only a limited number of solutions. Technically, Terwiesch and Xu (2008) characterize the equilibrium assuming that the value of each search follows a Gumbel distribution, whereas our results are derived for any continuous distribution. Additionally, we establish a stronger uniqueness result (except in rare corner cases): Whenever a symmetric pure-strategy equilibrium exists, the mixed-strategy symmetric equilibrium does not exist, and vice versa. Finally, the equilibrium characterization of parallel search in Proposition 1 provides a basis for a fair comparison with the sequential search model, which we discuss next.

4.2 Sequential Search

Our sequential search model described in Section 3.2 coincides with the model analyzed in Taylor (1995), who first observed that the dominant stopping strategy that solves the Bellman equation in (2) is always a threshold strategy. More specifically, we define a stopping time τ_i as a z_i -stopping strategy if $\tau_i = \min\{r = 1, \dots, R \mid X_{i,(r)} \geq z_i\}$, i.e., the supplier stops searching at the first instance when the maximum value among the focal supplier's search results exceeds z_i . We state this property in the following lemma.⁵

LEMMA 1. *Given a supplier i and any stopping times τ_j of suppliers $j, j \neq i$, there exists a unique threshold z_i given by $z_i = \min\{\hat{x} \in [0, 1] \mid A \cdot \mathbb{E}[(\Phi_i(X) - \Phi_i(\hat{x}))^+] \leq c\}$ such that supplier i 's best response is a z_i -stopping strategy.*

Lemma 1 establishes that the dominant strategy for any supplier is to adopt a threshold policy. Building on this lemma, Taylor (1995) proves the symmetry of the stopping strategy among all suppliers and characterizes the equilibrium stopping threshold z^* . To describe z^* , first note that the cumulative distribution function of supplier i 's best performance value under a z_i -stopping strategy can be expressed as

$$G(x; z_i, R) = \begin{cases} F(x)^R, & x \leq z_i \\ F(z_i)^R + [1 - F(z_i)^R] \cdot \frac{F(x) - F(z_i)}{1 - F(z_i)}, & x > z_i. \end{cases} \quad (4)$$

When $x \leq z_i$, the probability that the best performance value is less than x is the same as the probability that the performance value of all R searches is less than x , since the supplier will not stop before exhausting all R evaluation opportunities. When $x > z_i$, one must account for the additional event that the performance value from one search exceeds z_i , in which case this value becomes the best performance value. The unique equilibrium τ^* is then characterized in the following proposition.

PROPOSITION 2 (Proposition 2 of Taylor (1995)). *Consider a setting with n suppliers, where each supplier adopts a sequential search strategy and has at most R evaluation opportunities. Then, the unique symmetric*

⁵ This property is not explicitly proved in Taylor (1995). However, one can adapt the proof from Lippman and McCall (1976), which handles the special case when $\Phi_i(\hat{x}) = \hat{x}$, to prove this lemma. The proof is provided in the Appendix for completeness.

equilibrium is for all suppliers to enter and to search with a z^* -stopping strategy, where z^* is the unique solution to the equation

$$A \int_z^1 (G(x; z, R)^{n-1} - G(z; z, R)^{n-1}) dF(x) - c = 0. \quad (5)$$

That is, a supplier will stop if the value of a search exceeds z^* , and will continue otherwise.

Proposition 2 mirrors the findings of Taylor (1995). The key takeaway from Proposition 2 is that each supplier adopts the same threshold strategy in equilibrium. Specifically, each supplier continues searching until the value of a search exceeds the threshold z^* , where z^* is time-independent. In contrast to the case where suppliers search in parallel using a pure strategy, the total rounds of search conducted by each supplier is stochastic when sequential search strategies are adopted. This number follows a geometric distribution with success probability $1 - F(z^*)$ as $R \rightarrow \infty$. This property will be useful in our subsequent analysis.

5 Buyer's Preference for Sequential and Parallel Search

Building on the equilibrium results in Section 4, we proceed to analyze whether the buyer prefers suppliers adopting sequential or parallel search. To answer this question, in Section 5.1, we first consider a setting where both the number of suppliers (n) and evaluation opportunities per supplier (R) are fixed. This is potentially due to external or organizational constraints such that the buyer cannot easily alter those two parameters. For example, many established companies maintain an approved vendor list, under which they are authorized to procure only from pre-vetted suppliers. In some specialized areas, the approved vendor list is often limited, sometimes to as few as two or three suppliers (Dyer and Ouchi 1993, Langner and Seidel 2009, Mueller et al. 2016). Moreover, evaluation resources, such as personnel and equipments, cannot easily be changed in the short run. Of course, there are cases where the buyer may have more freedom to determine those parameters to further improve the innovation outcomes, we therefore analyze the buyer's optimal design problem (Section 5.2) where the buyer can jointly optimize n and R , as well as the suppliers' search strategies.

5.1 A Setting with Exogenous R and n

Given the number of suppliers (n) and evaluation opportunities per supplier (R), Theorem 1 below compares the buyer's profit when suppliers adopt sequential versus parallel search.

THEOREM 1. *Consider a setting with n suppliers and each has R evaluation opportunities.*

- (i) *The buyer's profit is always higher when suppliers adopt sequential search than parallel search, if there are infinitely many evaluation opportunities for each supplier. That is, as $R \rightarrow \infty$, then $\Pi_S > \Pi_P$, for any $c \in (0, A/n)$.*

- (ii) For a given number of evaluation opportunities, the buyer's profit is higher (resp. lower) when suppliers adopt sequential search than parallel search, if search costs are high (resp. low). That is, for any given $R < \infty$, there exists thresholds $0 \leq c_l < c_h \leq A/n$ such that: $\Pi_S > \Pi_P$ if $c \in (c_h, A/n)$ and $\Pi_S < \Pi_P$ if $c \in (0, c_l)$.

Theorem 1 shows that whether the buyer prefers suppliers adopting sequential or parallel search depends on two factors: the number of evaluation opportunities available to each supplier (R) and unit search cost (c). The results are summarized in Table 1. More specifically, the buyer prefers sequential search if each supplier is granted abundant evaluation opportunities or if the unit search cost is high. In contrast, the buyer prefers parallel search if the number of evaluation opportunities is limited *and* the unit search cost is low. This latter finding stands in sharp contrast to the previous results in the literature, which show that sequential search always dominates parallel search when time costs are zero due to its informational advantage arising from the adaptive nature (Loch et al. 2001). However, in our multi-agent competitive setting, we demonstrate that parallel search can still be preferred in the absence of time consideration. Therefore, it warrants a careful analysis of the economic drivers underlying sequential and parallel search.

5.1.1 Decomposing the Buyer's Profit Difference ($\Pi_S - \Pi_P$).

It is useful to understand what contributes to the buyer's profit difference ($\Pi_S - \Pi_P$) when suppliers adopt sequential versus parallel search. To proceed, we analyze a *hypothetical* scenario in which each of the n suppliers conducts $\tilde{m}$ searches in parallel, where $\tilde{m}$ has exactly the same distribution as the number of searches under sequential search. The key difference is that, in this hypothetical scenario, each supplier first decides the value of $\tilde{m}$ and then *blindly* conducts that many searches in parallel, whereas under sequential search the stopping decision is *adaptive*, with each supplier conducting one search at a time, and stopping whenever the value exceeds z^* . The profit associated with the hypothetical scenario is given by $\Pi_0 = \mathbb{E}[\max_{i=1, \dots, n} X_{i,(\tilde{m})}] - A$.

With the aid of this benchmark profit Π_0 , we can derive the following decomposition:

$$\Pi_S - \Pi_P = \underbrace{(\Pi_S - \Pi_0)}_{\text{Quality effect (+): due to informational advantage of sequential search}} + \underbrace{(\Pi_0 - \Pi_P)}_{\text{Quantity effect (+/-): due to difference in no. of searches across the two strategies}}. \quad (6)$$

Quality Effect: The first term ($\Pi_S - \Pi_0$) represents the buyer's profit difference between the cases when suppliers adopt sequential search and the hypothetical scenario. Because the only distinction between the two scenarios lies in whether suppliers can leverage performance feedback to conduct their searches more adaptively, the term ($\Pi_S - \Pi_0$) can be interpreted as the *informational advantage* of sequential search, which we refer to as *quality effect*. This benefit of sequential search has been well-established in the search literature (Loch et al. 2001), and it is always weakly positive (by Lemma 2 below).

LEMMA 2. *The quality effect, defined in (6), satisfies $\Pi_S - \Pi_0 \geq 0$.*

This search-specific feature is absent from conventional dynamic contest models based on effort plus noise (see, e.g., Mihm and Schlapp 2019).

Quantity Effect: Beyond its informational advantage of sequential search, the second term $(\Pi_0 - \Pi_P)$ on the right-hand side of (6) captures a more novel effect to the search literature and is key to understanding Theorem 1. Recall that this term captures the buyer's profit difference between the hypothetical scenario and the case where suppliers adopt parallel search. In both scenarios, suppliers conduct their searches in parallel. Therefore, the buyer's profit difference $(\Pi_0 - \Pi_P)$ can only be attributed to the difference in number of searches, which we refer to as *quantity effect*. This highlights that the performance feedback received during sequential search can affect suppliers' search incentives, as compared to parallel search.

The decomposition in (6), through the lens of quality and quantity effects, is instrumental to understanding Theorem 1, which we discuss next.

5.1.2 Understanding Theorem 1 Through Quality and Quantity Effects.

As discussed, the quality effect is always weakly positive, regardless of the values of the contextual parameters such as R or n . Therefore, whether $(\Pi_S - \Pi_P)$ is positive or negative hinges critically on the direction and magnitude of the quantity effect, which we examine next.

First, we consider the case of many evaluation opportunities, which corresponds to Theorem 1(i). Here, the quantity effect is also positive. This implies that when each supplier is granted abundant evaluation opportunities, the benefit of sequential search over parallel search is further reinforced by a quantity effect, besides the usual quality effect, as illustrated in Figure 1(a).

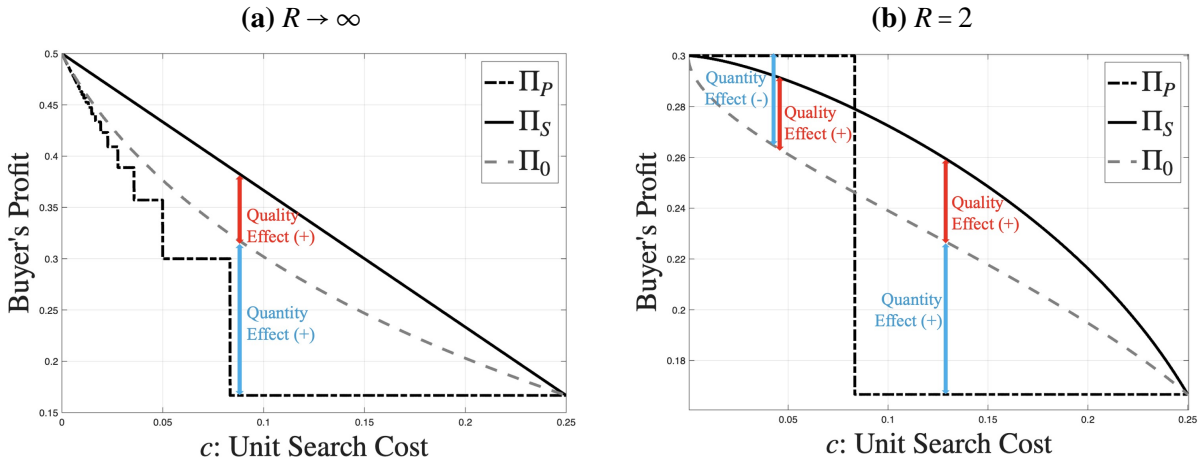


Figure 1 Buyer's profit as a function of search cost under different values of R

Notes: The quality- and quantity-effect decomposition is defined in (6), and the sign in parentheses indicates the direction of the effect. The parameters used are $A = 0.5$, $n = 2$, and $F \sim U[0, 1]$.

Next, we turn to the case with limited R , which corresponds to Theorem 1(ii). For ease of exposition, we focus on $R = 2$. When search costs are high, the decomposition closely mirrors the case of $R \rightarrow \infty$, with

both quantity and quality effects remaining positive (see Figure 1(b) under high search costs). The case of low search costs is perhaps more interesting. Here, the quantity effect is negative, as is evident by the negative $(\Pi_S - \Pi_0)$ in Figure 1(b) under low search costs. To understand the intuition, each supplier conducts two searches under parallel search, the maximum allowed when $R = 2$ (by Proposition 1). Under sequential search, by contrast, a supplier may stop after the first search if the realized outcome exceeds the threshold, leaving the second evaluation opportunity unused. This sheds light on why suppliers, on average, search more under parallel search when R is limited and search costs are low, substantiating the negative quantity effect in this regime. Moreover, this negative quantity effect is so pronounced that it cannot be offset by the positive quality effect (see again the red and blue arrows in Figure 1(b) under low search costs).

5.2 The Buyer's Optimal Design Problem

Section 5.1 analyzed a setting in which both the number of invited suppliers (n) and the number of evaluation opportunities (R) are fixed. However, buyers sometimes can alter those two parameters in order to further improve innovation outcomes. Therefore, we solve for the optimal design problem, with the buyer determining: (i) n , (ii) R , and (iii) the choice between sequential and parallel search.

As a preliminary step towards the joint optimization, Proposition 3 shows how the buyer's profit Π changes with respect to a single variable, R or n .

PROPOSITION 3. *The buyer's profit Π exhibits the following properties:*

- (i) *With the number of suppliers n fixed, the buyer's profits under both search strategies are increasing in R . That is, $\Pi_P(R)$ and $\Pi_S(R)$ increase in R .*
- (ii) *As the number of evaluation opportunities $R \rightarrow \infty$, the buyer's profit when suppliers adopt sequential search, $\Pi_S(n)$, decreases in n .*

Proposition 3(i) shows that, when the number of suppliers is fixed, granting more evaluation opportunities R improves the buyer's profit no matter suppliers adopt sequential or parallel search. This result is intuitive, as more evaluation opportunities give suppliers more flexibility, which in turn benefits the buyer. Building on this result, if the buyer can choose R without any restriction, it follows that the buyer's global optimum occurs as $R \rightarrow \infty$ (i.e., the buyer can afford to evaluate all submissions from each supplier). In addition, since sequential search dominates parallel search as $R \rightarrow \infty$ (see Figure 2(a)),⁶ the joint optimization reduces to understanding how $\Pi_S(n)$ changes with the number of suppliers n as $R \rightarrow \infty$.

Intuitively, the relationship between n and $\Pi_S(n)$ is ambiguous due to two countervailing forces. On one hand, increasing the number of suppliers intensifies competition and therefore, weakens search incentives. This effect is analyzed in Taylor (1995), who shows that the stopping threshold z^* decreases with n (see

⁶ Mathematically, as $R \rightarrow \infty$, we can show that $\Pi_S(n) > \Pi_P(n')$ for any $2 \leq n, n' < A/c$. This is a generalization of Theorem 1(i) which was formally proved in Theorem 1.

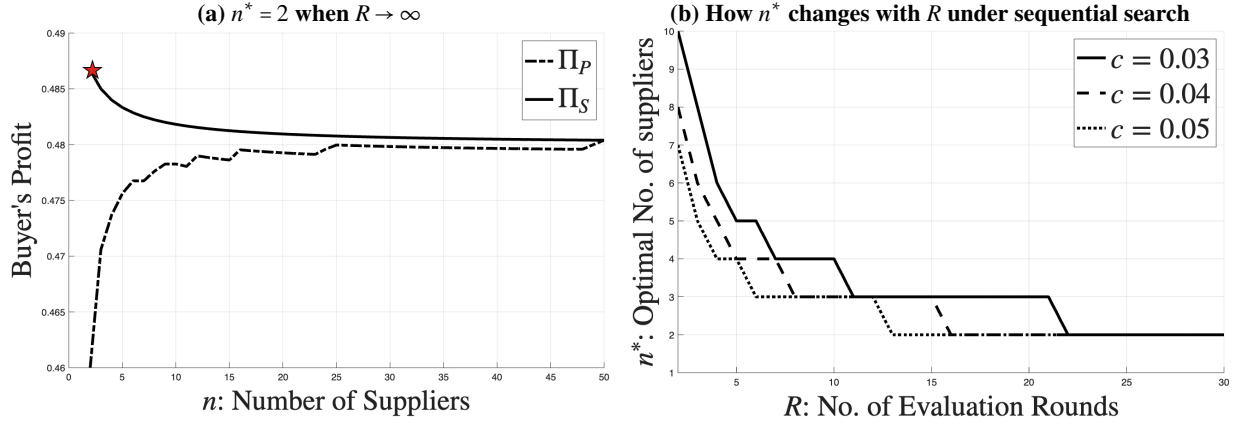


Figure 2 Optimal number of suppliers n^* as a function of R

Notes: The left panel shows how the buyer's profits, Π_S and Π_P , vary with n as $R \rightarrow \infty$. It is clear that Π_P is uniformly dominated by Π_S , and that the buyer's profit is maximized at $n = 2$. The right panel shows how n^* under sequential search changes with R for different values of c . As $R \rightarrow \infty$, the right panel shows $n^* \rightarrow 2$, in line with the left panel. The parameters used are $A = 0.5$, $F \sim U[0, 1]$, and, for the left panel, $c = 0.01$.

Equation A5), implying that suppliers tend to stop earlier as n increases. On the other hand, although each supplier searches less, the buyer can tap into a larger supplier pool as n increases. Overall, it is unclear which of these two forces dominates. Proposition 3(ii) provides a definitive answer by showing that $\Pi_S(n)$ decreases in n , as $R \rightarrow \infty$, indicating that the negative incentive effect dominates⁷.

Equipped with Proposition 3, the solution to the buyer's joint optimization problem is presented in the following theorem.

THEOREM 2. *When the buyer can freely choose the number of suppliers n and number of evaluation opportunities R , it is optimal to invite two suppliers (i.e., $n^* = 2$) and allow an unlimited number of evaluation opportunities (i.e., $R^* \rightarrow \infty$), under which all suppliers are required to search sequentially.*

The key takeaway of Theorem 2 is that: if the buyer can freely choose the number of suppliers n and number of evaluation opportunities R , the optimal policy is that he should engage only two suppliers, impose no limit on evaluation opportunities, and guide suppliers into sequential search. This result is derived based on the assumption that the buyer can afford to evaluate all submissions from each supplier (i.e., $R \rightarrow \infty$).

However, buyers may be resource-constrained in practice. Next, we analyze a setting where R is capped by an upper bound. Here, it then remains to determine (i) the number of suppliers n , and (ii) the choice between sequential and parallel search. Our extensive simulations show that, even under a cap on R , the buyer always prefers suppliers to adopt sequential search if she can optimally choose the number of suppliers under sequential and parallel search respectively.⁸ Moreover, the optimal number of suppliers n^* decreases in

⁷ Although Taylor (1995) characterizes how the stopping threshold z^* changes with n and provides intuition for this comparative-static effect, the paper does not establish a definitive result on how $\Pi_S(n)$ varies with n .

⁸ This is in contrast to our main results in Theorem 1 that when the number of suppliers is forced to be the same for both sequential and parallel search, then either can dominate the other depending on c and R .

R , converging to $n^* = 2$ in the limiting case as $R \rightarrow \infty$ (see Figure 2(b)). Beyond the role of R , Figure 2(b) further reveals that search costs c act as an additional moderating factor: higher search costs are associated with a smaller optimal supplier base.

Taken together, these results yield the following managerial insights for the buyer's optimal design problem. First, the buyer should steer suppliers toward sequential search by providing performance feedback and commit all available evaluation resources. Second, in order to configure an optimal supplier base, the buyer should always maintain a certain degree of restrictiveness than allow open entry. The optimal supplier base size should be tailored to the buyer's evaluation capacity and search cost, with higher evaluation capacity and higher search cost being associated with a smaller optimal number of suppliers.

The result on the optimal size of the supplier base has both theoretical and practical implications. From a theoretical perspective, it complements and enriches existing findings in the literature. Specifically, in a parallel search setting, Fullerton and McAfee (1999) show that the optimal supplier base consists of the two suppliers with the lowest and second-lowest costs, whereas Terwiesch and Xu (2008) show that the optimal number of suppliers is large (i.e., open entry) under symmetric suppliers. In a sequential search setting, although Taylor (1995) notes that inviting too many participants reduces each supplier's effort, the implication for the buyer's profit is ambiguous because the buyer can now access a larger supplier base. As a result, the optimal number of participants is not explicitly characterized in Taylor (1995). In contrast to the existing literature, we study a setting that encompasses both parallel and sequential search and show that the optimal number of suppliers is two if the buyer can afford to evaluate all suppliers' submissions (i.e., $R \rightarrow \infty$). Moreover, when R is limited, our numerical analysis shows that the optimal supplier base also features restricted entry and sheds light on how the optimal n^* varies with the buyer's evaluation capacity R and search cost c .

From a practical perspective, the result aligns closely with industrial procurement practices across a range of industries, including automotive (Langner and Seidel 2009, Mueller et al. 2016), electronics (McIvor 2001), defense (Roderick 2016), and flavor development (Van der Valk and Wynstra 2005), where supplier bases are often deliberately restricted, sometimes to as few as two suppliers (e.g., the well-known two-vendor policy adopted by Toyota, see Dyer and Ouchi 1993). In complex industrial projects such as automotive component development, the cost of developing a prototype can be high (i.e., high search costs). Our results showing that higher search costs are associated with a smaller optimal supplier base, therefore help shed light on the economic rationale underlying Toyota's two-vendor policy.

6 The Impact of Buyer's Evaluation Cost

6.1 A Setting with Exogenous R and n

To focus on the comparison between sequential and parallel search, our base model assumes that the buyer incurs no cost from evaluating suppliers' submitted solutions. In practice, however, evaluation can be costly

for the buyer. For example, in flavor development, each submitted flavor may need to be evaluated by the buyer's application specialists or even tested by end consumers. In this section, we define evaluation costs as the monetary costs associated with engaging these specialists or consumers. We then examine how such costs affect the comparison of the buyer's profits under the two search strategies.

More specifically, we introduce $c_e > 0$, the buyer's unit evaluation cost of one submitted solution. After accounting for evaluation costs, the buyer's profit when suppliers adopt parallel search strategies is given by $\Pi_p^e = \Pi_p - c_e \mathbb{E}[nm^*]$, where Π_p is the buyer's profit under parallel search without evaluation costs defined in (1). Similarly, accounting for evaluation costs, the buyer's profit when suppliers adopt sequential search strategies is given by $\Pi_s^e = \Pi_s - c_e \mathbb{E}[n\tau_i^*]$, where Π_s is the buyer's profit under sequential search without evaluation costs defined in (3), and τ_i^* is the equilibrium stopping time of supplier i . In this case, whether parallel search or sequential search is more profitable for the buyer is then determined by $\Delta^e = \Pi_s^e - \Pi_p^e = \Pi_s - \Pi_p - c_e (\mathbb{E}[n\tau_i^*] - \mathbb{E}[nm^*])$. Thus, relative to the base model, evaluation costs make the search strategy that induces less supplier search more attractive to the buyer. This effect can operate in either direction: when suppliers search more under sequential search, evaluation costs make sequential search less attractive; when they search less, evaluation costs make it more attractive.

Hence, after adding evaluation cost into the buyer's consideration, there is this trade-off: when suppliers conduct more searches, the expected value of their solutions increases, but the buyer also incurs higher evaluation costs. Because Δ_e decreases with c_e , when c_e is sufficiently small, the buyer's preference between sequential and parallel search remains the same as that characterized in Theorem 1 and summarized in Table 1. The more interesting case arises when c_e is sufficiently large. In this case, the buyer's preference reverses because the strategy that induces more supplier searches also generates a larger evaluation-cost burden. Theorem 3 formalizes this intuition.

THEOREM 3. *Consider a setting with n suppliers, each of whom has R search opportunities. When the unit evaluation cost is sufficiently large relative to the unit search cost, the buyer's preference between sequential and parallel search, as characterized in Theorem 1 and summarized in Table 1, is reversed. More specifically:*

- (i) *As $R \rightarrow \infty$, for any $c \in (0, A/n)$, we have $\Pi_s^e < \Pi_p^e$ for sufficiently large c_e .*
- (ii) *For any fixed $R < \infty$, let c_l and c_h denote the thresholds defined in Theorem 1(ii). For any $c \in (c_h, A/n)$, we have $\Pi_s^e < \Pi_p^e$ for sufficiently large c_e . In contrast, for any $c \in (0, c_l)$, we have $\Pi_p^e < \Pi_s^e$ for sufficiently large c_e .*

6.2 The Buyer's Optimal Design Problem with Evaluation Cost

In Section 5.2, we characterized the buyer's optimal design when n and R can be endogenized, assuming zero evaluation costs. In this section, after incorporating buyer evaluation costs into our model, we revisit this design problem and examine how the buyer's optimal choices change when each evaluated submission incurs a cost.

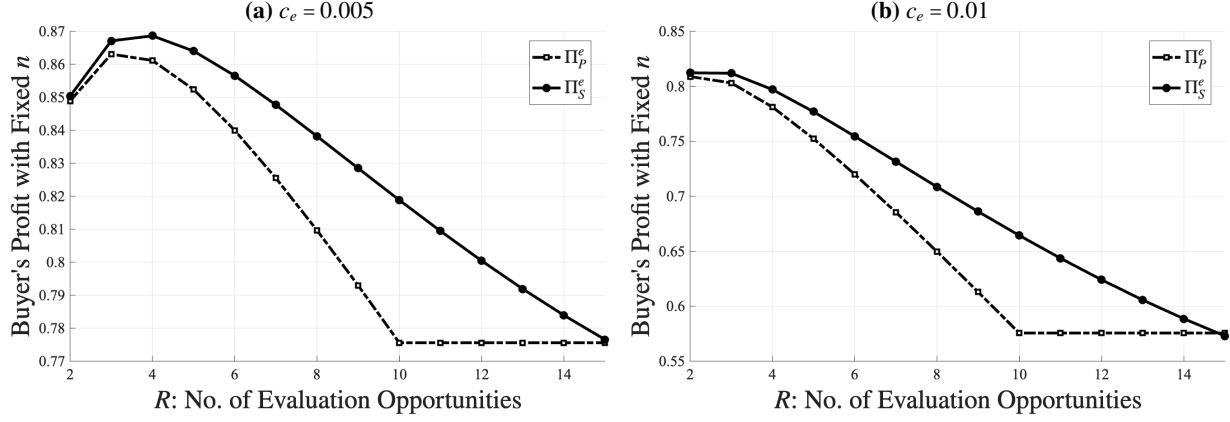


Figure 3 The buyer's profit as a function of R with fixed n

Notes: Both panels show how the buyer's profits, Π_S^e and Π_P^e , vary with R for $n = 4$. The parameters used are $A = 0.5$, $c = 0.01$ and $F \sim U[0, 1]$. The left panel sets $c_e = 0.005$, and the right panel sets $c_e = 0.01$.

As a preliminary step, we first examine how the buyer's profits change with an endogenized R for a fixed n . Recall that Proposition 3(i) shows that, when $c_e = 0$, the buyer's profits under both search strategies, $\Pi_P(R)$ and $\Pi_S(R)$, increase with R . Intuitively, a larger number of evaluation opportunities induces suppliers to search more, and because evaluation is costless, the resulting improvement in solution quality benefits the buyer. In Figure 3, we instead examine how the buyer's profit varies with the number of evaluation opportunities when evaluation costs are positive and the number of suppliers is fixed. Unlike the case with zero evaluation cost, our extensive numerical experiments show that the buyer's profit is generally no longer increasing in R . The reason is that a larger R typically induces more supplier searches, which in turn increases the buyer's total evaluation cost. Moreover, as c_e increases from 0.005 in the left panel to 0.01 in the right panel, the buyer's profit shifts from an increasing-then-decreasing pattern to a decreasing pattern in R , showing that higher evaluation costs make additional evaluations more costly for the buyer.

Next, we consider the buyer's optimal design when evaluation costs are taken into account. Recall that in Section 5.2, the buyer's profits under both search strategies increase with R , so the optimal design therefore involves sequential search with $R \rightarrow \infty$ and $n = 2$. Once evaluation costs are incorporated, however, the optimal design becomes more complex. In particular, unlike in the zero-evaluation-cost case, there is no single optimal design that applies uniformly across parameter values. Instead, the optimal design depends jointly on the unit search cost c and the unit evaluation cost c_e .

We first examine the case in which the unit search cost is relatively small, as shown in the left panel. In our numerical analysis in Figure 4(a), the optimal number of evaluation opportunities, R^* , generally decreases with c_e : as evaluation becomes more costly, the buyer prefers to restrict the total number of samples in order to save on evaluation costs. We also find that, as c_e increases, the optimal number of suppliers remains $n^* = 2$, and sequential search continues to dominate parallel search. To understand this pattern, note that when c_e is very small, the buyer optimally chooses a relatively large R^* . With a large R^* , sequential search tends to be

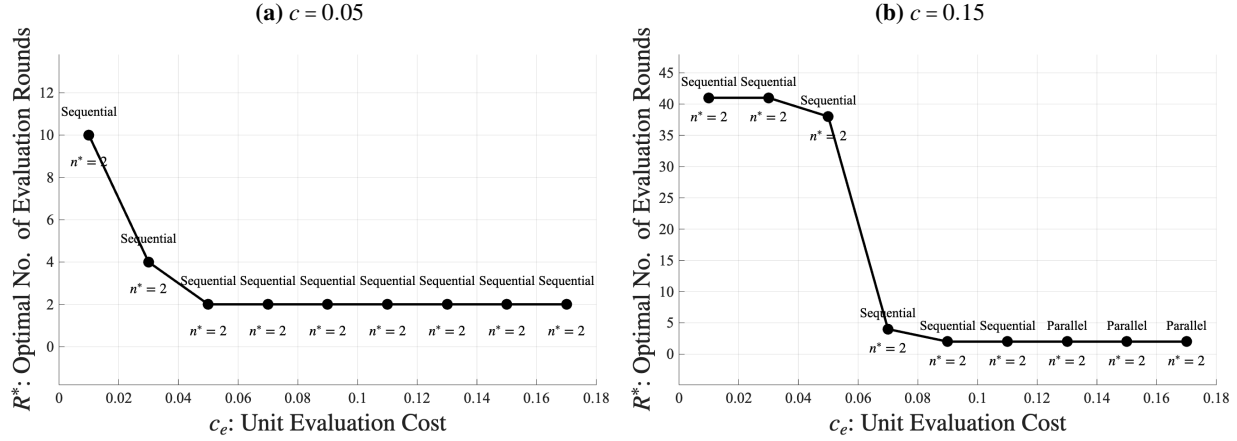


Figure 4 The buyer's optimal design with evaluation costs

Notes: Both panels show how the buyer's optimal design, R^* , n^* , as well as suppliers' search strategies, varies with c_e . The parameters used are $A = 0.5$ and $F \sim U[0, 1]$. The left panel sets $c = 0.05$, and the right panel sets $c = 0.15$.

more attractive because the buyer benefits more from both the quality and quantity effects. When c_e becomes larger, the optimal design eventually reduces to $R^* = 2$. Recall from Figure 1(b) that when the search cost is small and $R^* = 2$, without evaluation cost, parallel search dominates because suppliers search up to the upper bound and therefore generate more samples than under sequential search. However, given c_e is sufficiently large, this advantage becomes costly for the buyer, as the additional samples substantially increase evaluation costs. As a result, the buyer's preference shifts away from parallel search, and sequential search remains preferred even for large c_e .

When the unit search cost is large, as shown in the right panel, we observe a similar pattern for small values of c_e : R^* decreases as c_e increases, sequential search is preferred, and the optimal number of suppliers remains $n^* = 2$. However, the pattern changes when c_e becomes larger. When both the unit search cost and the unit evaluation cost are high, the optimal design switches from sequential search to parallel search as c_e becomes sufficiently large. This switch can be understood from Figure 1(b) as well. When $R^* = 2$ and the search cost is high, sequential search dominates because suppliers generate more than one sample, whereas under parallel search each supplier searches only once. However, when the evaluation cost becomes sufficiently large, the additional samples generated under sequential search become costly for the buyer. As a result, the buyer's preference reverses, and parallel search now becomes the optimal design.

In sum, when evaluation costs are taken into account, the main insights from Section 5.2 continue to hold in most cases: the buyer prefers sequential search and restricted entry. An exception arises when both the unit search cost and the unit evaluation cost are high, in which case the optimal search strategy switches from sequential search to parallel search.

7 Conclusion

Sequential and parallel search strategies represent two fundamental approaches to concept generation and NPD in general. To fully tap into the innovation potential of their supplier base, firms often delegate the

search process to multiple suppliers. In such a competitive, multi-agent setting, should the buyer prefer suppliers to adopt sequential or parallel search strategies?

To answer this question, we integrate the classical search model within a contest framework. When the number of suppliers and the number of evaluation opportunities per supplier are given (e.g., the buyer cannot easily change these parameters in the short run), our main findings are summarized in Table 1 in the Introduction. More specifically, we show that the buyer prefers sequential search when (i) unit search costs are high or (ii) when the maximum number of evaluation opportunities is large. We further uncover two benefits of sequential search: It incentivizes suppliers to conduct more searches (“quantity effect”) in a more efficient way (“quality effect”). In contrast, parallel search results in higher profits for the buyer when there are few evaluation opportunities and when search costs are low.

If the buyer has the freedom to decide on the number of supplier to invite and number of evaluation opportunities to offer per supplier, she should always steer suppliers toward sequential search and commit all available evaluation resources. Moreover, the optimal supplier base is always restrictive entry (as opposed to open entry). In particular, if the buyer can afford to evaluate all suppliers’ submissions without a cap on the number of evaluation opportunities, the optimal supplier base size is two.

We further extend our analysis to the case in which the buyer incurs a positive unit evaluation cost for each submission. We find that the main-model insights, derived under fixed numbers of suppliers and evaluation opportunities, continue to hold when evaluation costs are sufficiently small, but would be reversed when the evaluation cost is sufficiently large. As for the buyer’s optimal design problem in the presence of evaluation costs, we find that the buyer prefers restricted entry and sequential search in general, with the exception that when both the unit search cost and the unit evaluation cost are high, the optimal design switches from sequential search to parallel search.

There are several promising directions for future research. First, while this paper compares two canonical search strategies, future work could adopt a more normative perspective to characterize the optimal search strategies, which may involve a combination of the two. Second, in our model, the buyer steers suppliers toward sequential search by providing performance feedback on how well a proposed solution aligns with her preferences, whereas withholding feedback corresponds to a parallel search environment. Beyond performance feedback, the buyer could also offer directional feedback—specific guidance on how suppliers might improve their solutions in subsequent iterations. Exploring how such directional feedback influences suppliers’ search incentives presents an intriguing avenue for future study.

References

- Ales, L., S.-H. Cho, E. Körpeoğlu. 2017. Optimal award scheme in innovation tournaments. *Operations Research*, 65 (3), 693-702.
- Ales, L., S.-H. Cho, E. Körpeoğlu. 2019. Innovation and crowdsourcing contests. *Sharing Economy: Making Supply Meet Demand*, chap. 16. Springer, Cham, 379-406.

- Ales, L., S. H. Cho, E. Körpeoğlu. 2021. Innovation tournaments with multiple contributors. *Production and Operations Management*, 30 (6), 1772-1784.
- Aoyagi, M. 2010. Information feedback in a dynamic tournament. *Games and Economic Behavior*, 70 (2), 242-260.
- Bhaskaran, S., S. Erat. 2025. Optimal prototyping with noisy measurements. *Manufacturing & Service Operations Management*, 27 (1), 322-338.
- Bimpikis, K., S. Ehsani, M. Mostagir. 2019. Designing dynamic contests. *Operations Research*, 67 (2), 339-356.
- Bolton, Patrick, Christopher Harris. 1999. Strategic experimentation. *Econometrica*, 67 (2), 349-374.
- Candoğan, S., C. G. Körpeoğlu, C. S. Tang. 2025. Implications of team submissions in open innovation contests. *IMA Journal of Management Mathematics*, 36 (1), 1-19.
- Chen, B., B. Chen, D. Knyazev. 2022a. Information disclosure in dynamic research contests. *The RAND Journal of Economics*, 53 (1), 113-137.
- Chen, Z., J. Mihm, J. Schlapp. 2022b. Sourcing innovation: Integrated system or individual components? *Manufacturing & Service Operations Management*, 24 (2), 1056-1073.
- Chen, Z., J. Mihm, J. Schlapp. 2025. Sourcing innovation: When to own and when to control your supplier. *Manufacturing & Service Operations Management*, 27 (6), 1869-1888.
- Dahan, E., H. Mendelson. 2001. An extreme-value model of concept testing. *Management Science*, 47 (1), 102-116.
- Deng, S.-J., W. Elmaghraby. 2005. Supplier selection via tournaments. *Production and Operations Management*, 14 (2), 252-267.
- Dyer, J., W. Ouchi. 1993. Japanese-style partnerships: Giving companies a competitive edge. *MIT Sloan Management Review*, 35 (1), 51-63.
- Ederer, F. 2010. Feedback and motivation in dynamic tournaments. *Journal of Economics & Management Strategy*, 19 (3), 733-769.
- Ely, J. C., G. Georgiadis, S. Khorasani, L. Rayo. 2022. Optimal Feedback in Contests. *The Review of Economic Studies*, 90 (5), 2370-2394.
- Erat, S., S. Kavadias. 2008. Sequential testing of product designs: Implications for learning. *Management Science*, 54 (5), 956-968.
- Erat, S., V. Krishnan. 2012. Managing delegated search over design spaces. *Management Science*, 58 (3), 606-623.
- Fan, X., E. Körpeoğlu, C. Li, J. Liang. 2024. Innovative-product sourcing: Incentives and commitment. *Available at SSRN 4928116*, .
- Fullerton, Richard L, R Preston McAfee. 1999. Auctionin entry into tournaments. *Journal of Political Economy*, 107 (3), 573-605.
- Hu, M., L. Wang. 2021. Joint vs. separate crowdsourcing contests. *Management Science*, 67 (5), 2711-2728.

- Hu, Zhenyu, Wenjie Tang. 2021. Size matters, so does duration: The interplay between offer size and offer deadline. *Management Science*, 67 (8), 4935-4960.
- Hu, Zhenyu, Y. Xiao. 2024. Optimal dynamic mechanism under customer search. *Operations Research*, forthcoming.
- Huh, W. T., M. J. Kim, M. Lin. 2025. Uncertain search with knowledge transfer. *Management Science*, forthcoming.
- International Flavors & Fragrances Inc. 2024. International flavors & fragrances inc. annual report 2023. Tech. rep., International Flavors & Fragrances Inc. URL <https://www.iff.com/investors/financial-info/sec-filings>. Form 10-K for fiscal year ended December 31, 2023.
- Jiang, Z., Y. Huang, D. Beil. 2021. The role of problem specification in crowdsourcing contests for design problems: A theoretical and empirical analysis. *Manufacturing & Service Operations Management*, 23 (3), 637-656.
- Jiang, Z., Y. Huang, D. Beil. 2022. The role of feedback in dynamic crowdsourcing contests: A structural empirical analysis. *Management Science*, 68 (7), 4858-4877.
- Kavadias, S., J. Hutchison-Krupat. 2020. A framework for managing innovation. *Pushing the Boundaries: Frontiers in Impactful OR/OM Research*. INFORMS, 202-228.
- Khorasani, S., E. Körpeoğlu, V. Krishnan. 2024. Dynamic development contests. *Operations Research*, 72 (1), 43-59.
- Khorasani, S., L. Nittala, V. Krishnan. 2023. Screening in multistage contests. *Manufacturing & Service Operations Management*, 25 (6), 2249-2267.
- Körpeoğlu, C. G., E. Körpeoğlu, S. Tunç. 2021. Optimal duration of innovation contests. *Manufacturing & Service Operations Management*, 23 (3), 657-675.
- Körpeoğlu, E., S.-H. Cho. 2017. Incentives in contests with heterogeneous solvers. *Management Science*, 64 (6), 2709-2715.
- Körpeoğlu, E., C. G. Körpeoğlu, I. Hafalir. 2022. Parallel innovation contests. *Operations Research*, 70 (3), 1506-1530.
- Koslow, Scott, Sheila L Sasser, Edward A Riordan. 2006. Do marketers get the advertising they need or the advertising they deserve? agency views of how clients influence creativity. *Journal of Advertising*, 35 (3), 81-101.
- Langner, B., V. Seidel. 2009. Collaborative concept development using supplier competitions: Insights from the automotive industry. *Journal of Engineering and Technology Management*, 26 (1-2), 1-14.
- Liker, J., R. Kamath, S. Wasti, M. Nagamachi. 1996. Supplier involvement in automotive component design: Are there really large us japan differences? *Research Policy*, 25 (1), 59-89.
- Lin, M., C. S. Tang. 2025. Strategic probability setting and belief-driven search: A principal-agent framework. Working Paper.
- Lippman, S. A., J. J. McCall. 1976. The economics of job search: A survey. *Economic inquiry*, 14 (2), 155-189.
- Loch, C. H., C. Terwiesch, S. Thomke. 2001. Parallel and sequential testing of design alternatives. *Management Science*, 47 (5), 663-678.
- McCardle, K. F., I. Tsetlin, R. L. Winkler. 2018. When to abandon a research project and search for a new one. *Operations Research*, 66 (3), 799-813.

-
- McIvor, R. 2001. Lean supply: the design and cost reduction dimensions. *European Journal of Purchasing & Supply Management*, 7 (4), 227-242.
- Mihm, J., J. Schlapp. 2019. Sourcing innovation: On feedback in contests. *Management Science*, 65 (2), 559-576.
- Moldovanu, B., A. Sela. 2001. The optimal allocation of prizes in contests. *American Economic Review*, 91 (3), 542-558.
- Mueller, F., K. Stahl, F. Wachtler. 2016. Upstream relationships in the automotive industry: A contractual perspective. Working Paper.
- Oraiopoulos, N., S. Kavadias. 2014. The path-dependent nature of r&d search: Implications for (and from) competition. *Production and Operations Management*, 23 (8), 1450-1461.
- Rahmani, M., K. Ramachandran. 2021. Delegating innovation projects with deadline: Committed vs. flexible stopping. *Management Science*, 67 (10), 6317-6332.
- Roderick, J. V. 2016. Lightning versus thunder: The defense of the f-35 program. Master Thesis submitted to the Marine Corps Command and Staff College.
- Schlapp, J., G. Schumacher. 2022. Delegated concept testing in new product development. *Operations Research*, 70 (5), 2732-2748.
- Shalpegin, T., S. C. Sommer, Z. Wan. 2018. Collaborative prototyping of alternative designs under a target costing scheme. *Production and Operations Management*, 27 (3), 496-515.
- Sommer, S. C., C. H. Loch. 2004. Selectionism and learning in projects with complexity and unforeseeable uncertainty. *Management Science*, 50 (10), 1334-1347.
- Spence, C. 2015. Multisensory flavor perception. *Cell*, 161 (1), 24-35.
- Statista. 2024. Flavor and fragrances market worldwide - statistics & facts, May 28, 2024, <https://www.statista.com/topics/6300/flavor-and-fragrances-market-worldwide/#topicOverview>.
- Stouras, K., S. Erat, K. Lichtendahl Jr. 2025. Dueling contests and platform's coordinating role. *Management Science*, 71 (2), 1488-1503.
- Stouras, K., J. Hutchison-Krupat, R. Chao. 2022. The role of participation in innovation contests. *Management Science*, 68 (6), 4135-4150.
- Swift, J. 2014. Unilever turns to non-roster agencies for global axe pitch, November 12, 2014, <https://www.campaignlive.co.uk/article/unilever-turns-non-roster-agencies-global-axe-pitch/1321594>.
- Taylor, C. 1995. Digging for golden carrots: An analysis of research tournaments. *American Economic Review*, 85 (4), 872-890.
- Terwiesch, C., C. H. Loch. 2004. Collaborative prototyping and the pricing of custom-designed products. *Management Science*, 50 (2), 145-158.
- Terwiesch, C., Y. Xu. 2008. Innovation contests, open innovation, and multiagent problem solving. *Management Science*, 54 (9), 1529-1543.

- Thomke, S. 2003a. *Teaching Note: Bush Boake Allen*. HBS Case 5-603-115, Harvard Business School, Boston.
- Thomke, S., A. Nimgrade. 2000. *Bush Boake Allen*. HBS Case 9-601-061 (Revised June 3, 2010), Harvard Business School, Boston.
- Thomke, S. H. 2003b. *Experimentation matters: unlocking the potential of new technologies for innovation*. Harvard Business Press, Boston, MA.
- Ulrich, K. T., S. D. Eppinger. 2012. *Product design and development*. McGraw-Hill, New York, NY.
- Van der Valk, W., F. Wynstra. 2005. Supplier involvement in new product development in the food industry. *Industrial Marketing Management*, 34 (7), 681-694.
- Weitzman, M. 1979. Optimal search for the best alternative. *Econometrica*, 47 (3), 641-654.
- Wooten, J. O., K. T. Ulrich. 2017. Idea generation and the role of feedback: Evidence from field experiments with innovation tournaments. *Production and Operations Management*, 26 (1), 80-99.
- Xiao, Y., Z. Hu, S. Wang. 2024. Information design of a delegated search. *Available at SSRN 4249165*, .
- Zorc, S., I. Tsetlin. 2020. Deadlines, offer timing, and the search for alternatives. *Operations Research*, 68 (3), 927-948.
- Zorc, S., I. Tsetlin, S. Hasija, S. E Chick. 2025. The when and how of delegated search. *Operations Research*, 73 (1), 461-482.

Appendix

A Proofs

Proof of Proposition 1. As a preparation step to prove Proposition 1, we first simplify the suppliers' payoff function by providing a closed-form expression for a supplier's winning probability given a parallel search strategy profile $m = (m_1, \dots, m_n)$ in the lemma below.

LEMMA A1. *Given a parallel search strategy profile $m = (m_1, \dots, m_n)$, the winning probability of supplier i is given by*

$$\mathbb{P}(X_{i,(m_i)} > X_{j,(m_j)}, \quad j \neq i) = \frac{m_i}{\sum_{j=1}^n m_j}.$$

Proof of Lemma A1. Note that the probability density function of the $X_{i,(m_i)}$, denote as $f_{X_{i,(m_i)}}(x)$, for any $i = 1, \dots, n$, can be expressed as $f_{X_{i,(m_i)}}(x) = m_i \cdot F(x)^{m_i-1} \cdot f(x)$. Let $f_{\max_{j \neq i} X_{j,(m_j)}}(x)$ denote the probability density function of $\max_{j \neq i} X_{j,(m_j)}$, i.e., $f_{\max_{j \neq i} X_{j,(m_j)}}(x) = \sum_{j \neq i} m_j \cdot F(x)^{\sum_{j \neq i} m_j-1} \cdot f(x)$. We then have

$$\begin{aligned} \mathbb{P}(X_{i,(m_i)} > X_{j,(m_j)}, \quad j \neq i) &= \int_0^1 \int_y^1 f_{X_{i,(m_i)}}(x) \cdot f_{\max_{j \neq i} X_{j,(m_j)}}(y) dx dy \\ &= \int_0^1 \int_y^1 m_i \cdot F(x)^{m_i-1} \cdot f(x) \cdot \sum_{j \neq i} m_j \cdot F(y)^{\sum_{j \neq i} m_j-1} \cdot f(y) dx dy \\ &= \int_0^1 \left(\int_y^1 m_i \cdot F(x)^{m_i-1} \cdot f(x) dx \right) \cdot \sum_{j \neq i} m_j \cdot F(y)^{\sum_{j \neq i} m_j-1} \cdot f(y) dy \\ &= \int_0^1 (F(x)^{m_i} \Big|_y^1) \cdot \sum_{j \neq i} m_j \cdot F(y)^{\sum_{j \neq i} m_j-1} \cdot f(y) dy \\ &= \int_0^1 [1 - F(y)^{m_i}] \cdot \sum_{j \neq i} m_j \cdot F(y)^{\sum_{j \neq i} m_j-1} \cdot f(y) dy \\ &= (F(y)^{\sum_{j \neq i} m_j} \Big|_0^1) - \left(\frac{\sum_{j \neq i} m_j}{\sum_{j=1}^n m_j} F(y)^{\sum_{j=1}^n m_j} \Big|_0^1 \right) \\ &= 1 - \frac{\sum_{j \neq i} m_j}{\sum_{j=1}^n m_j} \\ &= \frac{m_i}{\sum_{j=1}^n m_j}. \end{aligned}$$

□

Lemma A1 shows that the winning probability of a supplier only depends on the number of searches conducted by all suppliers and it is independent of the distribution of the performance value. Based on Lemma A1, supplier i 's expected payoff given all his $n - 1$ opponents are doing m searches could be further simplified into

$$\begin{aligned} \pi_i(m_i | m) &= A \cdot \mathbb{P}(X_{i,(m_i)} > X_{j,(m)}, \quad j \neq i) - cm_i \\ &= A \cdot \frac{m_i}{m_i + (n-1)m} - cm_i. \end{aligned}$$

Our next lemma shows that $\pi_i(m_i | m)$ is strictly concave in m_i , which is again essential for the proof of Proposition 1.

LEMMA A2. $\pi_i(m_i | m)$ is strictly concave in m_i , i.e., $\pi_i(m_i + 1 | m) - \pi_i(m_i | m)$ is strictly decreasing in m_i .

Proof of Lemma A2. First observe that $\pi_i(m_i | m)$ is strictly concave in the integer domain if and only if its natural extension $\pi_i(m_i | m), m_i \in \mathbb{R}$ is strictly concave in real numbers. Hence, in the following we treat m_i as a real number and take the first order derivative of $\pi_i(m_i | m)$ with respect to m_i ,

$$\begin{aligned} \frac{\partial \pi_i(m_i | m)}{\partial m_i} &= A \cdot \frac{m_i + (n-1)m - m_i}{[m_i + m(n-1)]^2} - c \\ &= A \cdot \frac{m(n-1)}{[m_i + m(n-1)]^2} - c, \end{aligned}$$

which is strictly decreasing in m_i . Hence, $\pi_i(m_i | m)$ is strictly concave in m_i for $m_i = 1, \dots, R$. $\square$

We now prove Proposition 1 by first defining the equilibrium. By definition, a symmetric strategy $m_i = m, i = 1, \dots, n$ is a Nash Equilibrium if $\pi_i(s | m) - \pi_i(m | m) \leq 0$ for any $s = 1, \dots, R$ and $i = 1, \dots, n$. From the concavity of $\pi_i(m_i | m)$ proved in Lemma A2, for any $s > m$, we have

$$\begin{aligned} \pi_i(s | m) - \pi_i(m | m) &= \sum_{t=0}^{s-m-1} [\pi_i(s-t | m) - \pi_i(s-t-1 | m)] \\ &\leq (s-m) \cdot [\pi_i(m+1 | m) - \pi_i(m | m)]. \end{aligned} \quad (7)$$

Similarly for any $s < m$, we have

$$\begin{aligned} \pi_i(s | m) - \pi_i(m | m) &= \sum_{t=0}^{m-s-1} [\pi_i(s+t | m) - \pi_i(s+t+1 | m)] \\ &\leq (m-s) \cdot [\pi_i(m-1 | m) - \pi_i(m | m)]. \end{aligned} \quad (8)$$

Let us first derive the necessary and sufficient conditions for a strategy $1 < m < R$ being a Nash Equilibrium. Due to (7) and (8), if $1 \leq m \leq R$, $m_i = m, i = 1, \dots, n$ is a Nash Equilibrium if and only if

$$\pi_i(m+1 | m) - \pi_i(m | m) \leq 0 \quad (9)$$

$$\pi_i(m-1 | m) - \pi_i(m | m) \leq 0. \quad (10)$$

Similarly, $m_i = 1, i = 1, \dots, n$ is a Nash Equilibrium if and only if

$$\pi_i(2 | 1) - \pi_i(1 | 1) \leq 0. \quad (11)$$

And $m_i = R, i = 1, \dots, n$ is a Nash Equilibrium if and only if

$$\pi_i(R-1 | R) - \pi_i(R | R) \leq 0. \quad (12)$$

By Lemma A1, we could write (9) explicitly as

$$\begin{aligned} &\pi_i(m+1 | m) - \pi_i(m | m) \\ &= A \cdot \mathbb{P}(X_{i,(m+1)} > X_{j,(m)}, j \neq i) - c(m+1) - A \cdot \mathbb{P}(X_{i,(m)} > X_{j,(m)}, j \neq i) + cm \\ &= A \cdot \left[\frac{m+1}{m+1+(n-1)m} - \frac{m}{m+(n-1)m} \right] - c \leq 0, \end{aligned}$$

which is equivalent to $m \geq m_0 - 1/n$. Inequality (10) could be written as

$$\begin{aligned} & \pi_i(m-1 | m) - \pi_i(m | m) \\ &= A \cdot \mathbb{P}(X_{i,(m-1)} > X_{j,(m)}, j \neq i) - c(m-1) - A \cdot \mathbb{P}(X_{i,(m)} > X_{j,(m)}, j \neq i) + cm \\ &= A \cdot \left[\frac{m-1}{m-1+(n-1)m} - \frac{m}{m+(n-1)m} \right] + c \leq 0, \end{aligned}$$

which is equivalent to $m \leq m_0 + 1/n$. Similarly, inequality (11) is equivalent to $1 \geq m_0 - 1/n$. On the other hand, inequality (12) is equivalent to $R \leq m_0 + 1/n$.

Let us start the proof for part (i). When $R \geq \lceil m_0 \rceil$, if $[m_0 - 1/n, m_0 + 1/n]$ contains an integer s , it must be either $\lfloor m_0 \rfloor$ or $\lceil m_0 \rceil$. Also note by our assumption that $A > cn$, we must have $m_0 - 1/n > n \cdot (n-1)/n^2 - 1/n \geq 0$, for any $n \geq 2$. Hence we must have $1 \leq s$. By the necessary and sufficient conditions above, s must be a symmetric pure-strategy Nash Equilibrium. Moreover, if s is the unique integer inside the interval, then $m_i = s$ is the unique symmetric pure-strategy Nash Equilibrium. Part (i) thus follows.

For part (iii), when $R < \lceil m_0 \rceil$, we must have $R < m_0 < m_0 + 1/n$ and it is impossible that the interval $[m_0 - 1/n, m_0 + 1/n]$ contains any integer that is less than R . Hence, when $R < \lceil m_0 \rceil$, $m_i = R, i = 1, \dots, n$ will be the unique symmetric Nash equilibrium.

Next we continue with the proof for part (ii) by characterizing the unique symmetric mixed strategy Nash Equilibrium when $R \geq \lceil m_0 \rceil$ and the interval $[m_0 - 1/n, m_0 + 1/n]$ does not contain an integer, i.e., when a symmetric pure-strategy Nash Equilibrium does not exist. We first show in the following lemma that in any mixed-strategy equilibrium, all suppliers must be mixing between two consecutive integers.

LEMMA A3. *In any symmetric mixed strategy equilibrium, each supplier's equilibrium number of searches $\tilde{m}$ is characterized by*

$$\tilde{m} = \begin{cases} r, & \text{with probability } p \\ r+1, & \text{with probability } 1-p, \end{cases}$$

for some $r \in \{1, \dots, R-1\}$ and $p \in (0, 1)$.

Proof of Lemma A3. Suppose $\tilde{m}_j$ is the random number of searches conducted by supplier j , $j = 1, \dots, n$, which will all follow an identical distribution in a symmetric mixed strategy equilibrium. The expected payoff of supplier i conducting m_i searches given that his opponents' strategy following $\tilde{m}_{-i}$ is then $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m_i | \tilde{m}_{-i})]$.

For any given realization of the random numbers of searches done by supplier i 's opponents $m_{-i} = (m_1, \dots, m_{i-1}, m_{i+1}, \dots, m_n)$, we have $\pi_i(m_i | m_{-i}) = A \cdot m_i / \sum_{j=1}^n m_j - cm_i$, which is strictly concave in m_i by Lemma A2. As strict concavity is preserved under expectation, $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m_i | \tilde{m}_{-i})]$ is also strictly concave in m_i .

Suppose that in equilibrium $\tilde{m}_i$ is distributed in a way such that $\mathbb{P}(\tilde{m}_i = r_k) = p_k$, for some $r_1, \dots, r_K \in \{1, \dots, R\}$. Then it must be true that $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r_1 | \tilde{m}_{-i})] = \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r_2 | \tilde{m}_{-i})] = \dots = \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r_K | \tilde{m}_{-i})] \geq \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m | \tilde{m}_{-i})]$, for all $m \in \{1, \dots, R\}, m \neq r_1, \dots, r_K$. Otherwise if $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r_k | \tilde{m}_{-i})]$ has a strictly larger value

for some k or $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m | \tilde{m}_{-i})]$ has a strictly larger value than $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r_1 | \tilde{m}_{-i})] = \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r_2 | \tilde{m}_{-i})] = \dots = \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r_K | \tilde{m}_{-i})]$ for some $m \neq r_1, \dots, r_K$, conducting $\tilde{m}_i$ number of searches is no longer a best response to $\tilde{m}_{-i}$.

By definition of strict concavity, $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m_i + 1 | \tilde{m}_{-i})] - \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m_i | \tilde{m}_{-i})]$ is strictly decreasing in m_i . Hence, there could be at most one $r \in \{1, \dots, R-1\}$ such that $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r | \tilde{m}_{-i})] = \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(r+1 | \tilde{m}_{-i})] > \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m | \tilde{m}_{-i})]$ for all $m \neq r, r+1$, which proves Lemma A3. $\square$

Based on Lemma A3, we can now focus on the symmetric mixed strategies of the form $\mathbb{P}(\tilde{m}_j = m) = p$ and $\mathbb{P}(\tilde{m}_j = m+1) = 1-p$ for all $j = 1, \dots, n$ and for some $m = 1, \dots, R-1$. In this case, we can explicitly write out the expected payoff of supplier i conducting m_i searches given the opponents' strategies $\tilde{m}_{-i}$ as:

$$\begin{aligned} & \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m_i | \tilde{m}_{-i})] \\ &= A \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-k-1} \cdot \mathbb{P}[(X_{(m_i)} > X_{j,(m)}, j=1, \dots, k) \cap (X_{(m_i)} > X_{j,(m+1)}, j=k+1, \dots, n-1)] - cm_i \\ &= A \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-k-1} \frac{m_i}{m_i + mk + (m+1)(n-k-1)} - cm_i. \end{aligned}$$

Here, the index k indicates that there are exactly k suppliers whose realized equilibrium number of searches being m . In addition, we use above the generic notation $X_{(m_i)}$ to indicate supplier i 's best performance value whereas $X_{j,(m)}$, $j=1, \dots, k$ denote k i.i.d. samples of best performance value from any k suppliers. We further let $H_i(m, p) = \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m+1 | \tilde{m}_{-i})] - \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m | \tilde{m}_{-i})]$ be the marginal expected profit of supplier i by conducting one more search, given his opponents are mixing between m and $m+1$ with probability p and $1-p$ respectively.

Again, for the mixed strategy profile $(\tilde{m}_1, \dots, \tilde{m}_n)$ to be a symmetric Nash Equilibrium, it is necessary that $H_i(m, p) = 0$ for all $i = 1, \dots, n$. This condition is also sufficient due to the strict concavity of $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m_i | \tilde{m}_{-i})]$. The following lemma characterizes the properties of $H_i(m, p)$.

LEMMA A4. $H_i(m, p)$ is strictly decreasing in m and strictly increasing in p .

Proof of Lemma A4. Using the expression for $\mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m_i | \tilde{m}_{-i})]$, we can write $H_i(m, p)$ explicitly as

$$\begin{aligned} H_i(m, p) &= \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m+1 | \tilde{m}_{-i})] - \mathbb{E}_{\tilde{m}_{-i}}[\pi_i(m | \tilde{m}_{-i})] \\ &= A \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-k-1} \left[\frac{m+1}{m+1+mk+(m+1)(n-k-1)} - \frac{m}{m+mk+(m+1)(n-k-1)} \right] - c \\ &= A \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-k-1} \left(\frac{m+1}{mn+n-k} - \frac{m}{mn+n-k-1} \right) - c. \end{aligned}$$

Note that $H_i(m, p)$ is strictly decreasing in the integer domain if and only if its natural continuous extension is also strictly decreasing. Hence, in the following we treat m as a continuous variable. Taking first order derivative of $H_i(m, p)$ with respect to m ,

$$\begin{aligned} \frac{\partial H_i(m, p)}{\partial m} &= A \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-k-1} \left[\frac{mn+n-k-(m+1)n}{(mn+n-k)^2} - \frac{mn+n-k-1-mn}{(mn+n-k-1)^2} \right] \\ &= A \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} p^k (1-p)^{n-k-1} \left[\frac{-k}{(mn+n-k)^2} + \frac{-(n-k-1)}{(mn+n-k-1)^2} \right], \end{aligned}$$

which is smaller than 0. Hence $H_i(m, p)$ is strictly decreasing in m .

Taking first order derivative of $H_i(m, p)$ with respect to p ,

$$\frac{\partial H_i(m, p)}{\partial p} = A \cdot \sum_{k=0}^{n-1} \binom{n-1}{k} [kp^{k-1}(1-p)^{n-k-1} + (n-k-1)p^k(1-p)^{n-k-2}] \left(\frac{m+1}{mn+n-k} - \frac{m}{mn+n-k-1} \right),$$

in which the term $kp^{k-1}(1-p)^{n-k-1} + (n-k-1)p^k(1-p)^{n-k-2}$ is always positive for any $k = 0, \dots, n-1$.

The last term can be computed as

$$\begin{aligned} \frac{m+1}{mn+n-k} - \frac{m}{mn+n-k-1} &= \frac{(mn+n-k)(m+1) - (m+1)m - m(mn+n-k)}{(mn+n-k)(mn+n-k-1)} \\ &= \frac{m(n-1) + (n-k-1)}{(mn+n-k)(mn+n-k-1)} > 0. \end{aligned}$$

Thus $H_i(m, p)$ is strictly increasing in p . $\square$

Now for part (ii), when the interval $[m_0 - 1/n, m_0 + 1/n]$ does not contain an integer, we first show that this implies $H_i(\lfloor m_0 \rfloor, 0) < 0$ and $H_i(\lfloor m_0 \rfloor, 1) > 0$. Indeed, $H_i(\lfloor m_0 \rfloor, 0) < 0$ is equivalent to

$$\begin{aligned} &A \cdot \left(\frac{\lfloor m_0 \rfloor + 1}{\lfloor m_0 \rfloor + 1 + (n-1)(\lfloor m_0 \rfloor + 1)} - \frac{\lfloor m_0 \rfloor}{\lfloor m_0 \rfloor + (n-1)(\lfloor m_0 \rfloor + 1)} \right) - c < 0 \\ \Leftrightarrow &A \cdot \left(\frac{1}{n} - \frac{\lfloor m_0 \rfloor}{n\lfloor m_0 \rfloor + (n-1)} \right) < c \\ \Leftrightarrow &\lfloor m_0 \rfloor + 1 > m_0 + \frac{1}{n} \\ \Leftrightarrow &\lfloor m_0 \rfloor > m_0 + \frac{1}{n}, \end{aligned}$$

which must hold when the interval $[m_0 - 1/n, m_0 + 1/n]$ does not contain any integer.

Similarly, $H_i(\lfloor m_0 \rfloor, 1) > 0$ is equivalent to

$$\begin{aligned} &A \cdot \left(\frac{\lfloor m_0 \rfloor + 1}{\lfloor m_0 \rfloor + 1 + (n-1)\lfloor m_0 \rfloor} - \frac{\lfloor m_0 \rfloor}{\lfloor m_0 \rfloor + (n-1)\lfloor m_0 \rfloor} \right) - c < 0 \\ \Leftrightarrow &A \cdot \left(\frac{\lfloor m_0 \rfloor + 1}{\lfloor m_0 \rfloor + 1 + (n-1)\lfloor m_0 \rfloor} - \frac{1}{n} \right) > c \\ \Leftrightarrow &\lfloor m_0 \rfloor < m_0 - \frac{1}{n}, \end{aligned}$$

which also holds due to the non-existence of integer in $[m_0 - 1/n, m_0 + 1/n]$.

Finally, since $H_i(\lfloor m_0 \rfloor, p)$ is strictly increasing in p , there exists a unique p^* as characterized in the statement of part (ii) that solves $H_i(\lfloor m_0 \rfloor, p) = 0$. This shows that the mixed strategy characterized by the pair $(\lfloor m_0 \rfloor, p)$ is indeed a Nash Equilibrium. In addition, we have $H_i(m, 0) - H_i(m+1, 1) = [\pi_i(m+1 | m+1) - \pi_i(m | m+1)] - [\pi_i(m+2 | m+1) - \pi_i(m+1 | m+1)]$, which is strictly greater than 0 by concavity of $\pi_i(m_i | m)$ proved in Lemma A2. Hence, $H_i(m, p) \geq H_i(m, 0) > H_i(\lfloor m_0 \rfloor, 1) > 0$ for all $m < \lfloor m_0 \rfloor, p \in [0, 1]$ and $H_i(m, p) \leq H_i(m, 1) < H_i(\lfloor m_0 \rfloor, 0) < 0$ for all $m > \lfloor m_0 \rfloor, p \in [0, 1]$. Therefore, the equilibrium we found is also unique. Note that when the interval $[m_0 - 1/n, m_0 + 1/n]$ doesn't contain an integer, we have $\lfloor m_0 \rfloor = \lceil m_0 \rceil - 1 >$

$m_0 + 1/n > n \cdot (n-1)/n^2 + 1/n - 1 = 0$, that is to say, suppliers will only mix between two consecutive positive integers. $\square$

Proof of Lemma 1. To simplify the notation, let $Z_{i,r}(\hat{x}_{i,r})$ denote the value-to-go if supplier i chooses to conduct a search in round r , i.e., $Z_{i,r}(\hat{x}_{i,r}) = -c + \mathbb{E}[V_{i,r+1}(\max\{\hat{x}_{i,r}, X_{i,r}\} \mid \tau_{-i})]$, $r = 1, \dots, R$. Define $z_{i,r} = \min\{\hat{x}_{i,r} : A \cdot \Phi_i(\hat{x}_{i,r}) - Z_{i,r}(\hat{x}_{i,r}) \geq 0\}$. We want to prove $z_{i,r}$ is indeed an optimal stopping threshold for supplier i in round r that solves equation (2), by proving that $A \cdot \Phi_i(\hat{x}_{i,r}) - Z_{i,r}(\hat{x}_{i,r})$ is increasing in $\hat{x}_{i,r}$ in any round r . The proof is done by backward induction. For $r = R$, we have

$$\begin{aligned} A \cdot \Phi_i(\hat{x}_{i,R}) - Z_{i,R}(\hat{x}_{i,R}) &= A \cdot \Phi_i(\hat{x}_{i,R}) + c - \mathbb{E}[V_{i,R+1}(\max\{\hat{x}_{i,R}, X_{i,R}\} \mid \tau_{-i})] \\ &= A \cdot \Phi_i(\hat{x}_{i,R}) + c - A \cdot \mathbb{E}[\Phi_i(\max\{\hat{x}_{i,R}, X_{i,R}\})] \\ &= A \cdot \Phi_i(\hat{x}_{i,R}) - A \cdot \mathbb{E}[\max\{\Phi_i(\hat{x}_{i,R}), \Phi_i(X_{i,R})\}] + c \\ &= c - A \cdot \mathbb{E}[(\Phi_i(X_{i,R}) - \Phi_i(\hat{x}_{i,R}))^+], \end{aligned}$$

which is increasing in $\hat{x}_{i,R}$. The third equality above is based on the increasing property of $\Phi_i(\cdot)$.

Suppose that in any round r , $A \cdot \Phi_i(\hat{x}_{i,r}) - Z_{i,r}(\hat{x}_{i,r})$ is increasing in $\hat{x}_{i,r}$. Then for round $r-1$, we have

$$\begin{aligned} A \cdot \Phi_i(\hat{x}_{i,r-1}) - Z_{i,r-1}(\hat{x}_{i,r-1}) &= A \cdot \Phi_i(\hat{x}_{i,r-1}) + c - \mathbb{E}[V_{i,r}(\max\{\hat{x}_{i,r-1}, X_{i,r-1}\} \mid \tau_{-i})] \\ &= A \cdot \Phi_i(\hat{x}_{i,r-1}) - \mathbb{E}[\max\{V_{i,r}(\hat{x}_{i,r-1} \mid \tau_{-i}), V_{i,r}(X_{i,r-1} \mid \tau_{-i})\}] + c \\ &= A \cdot \Phi_i(\hat{x}_{i,r-1}) - V_{i,r}(\hat{x}_{i,r-1} \mid \tau_{-i}) - \mathbb{E}([V_{i,r}(X_{i,r-1} \mid \tau_{-i}) - V_{i,r}(\hat{x}_{i,r-1} \mid \tau_{-i})]^+) + c \\ &= A \cdot \Phi_i(\hat{x}_{i,r-1}) - \max\{A \cdot \Phi_i(\hat{x}_{i,r-1}), Z_{i,r}(\hat{x}_{i,r-1})\} - \mathbb{E}([V_{i,r}(X_{i,r-1} \mid \tau_{-i}) - V_{i,r}(\hat{x}_{i,r-1} \mid \tau_{-i})]^+) + c \\ &= \min\{0, A \cdot \Phi_i(\hat{x}_{i,r-1}) - Z_{i,r}(\hat{x}_{i,r-1})\} - \mathbb{E}([V_{i,r}(X_{i,r-1} \mid \tau_{-i}) - V_{i,r}(\hat{x}_{i,r-1} \mid \tau_{-i})]^+) + c \end{aligned}$$

The second equality above is due to the increasing property of $V_{i,r}(\cdot)$, which can be easily established via induction. In the last equality, the term $A \cdot \Phi_i(\hat{x}_{i,r-1}) - Z_{i,r}(\hat{x}_{i,r-1})$ is increasing in $\hat{x}_{i,r-1}$ by our induction hypothesis, and the term $-\mathbb{E}([V_{i,r}(X_{i,r-1} \mid \tau_{-i}) - V_{i,r}(\hat{x}_{i,r-1} \mid \tau_{-i})]^+)$ is also increasing in $\hat{x}_{i,r-1}$ due to the increasing property of $V_{i,r}(\cdot)$. As a result, $A \cdot \Phi_i(\hat{x}_{i,r-1}) - Z_{i,r-1}(\hat{x}_{i,r-1})$ is also increasing in $\hat{x}_{i,r-1}$. By the definition of $z_{i,r}$, it is then optimal for the supplier i to stop in round r if and only if $\hat{x}_{i,r} \geq z_{i,r}$.

Next we show that the stopping threshold $z_{i,r}$ is in fact stationary, i.e., $z_{i,r} = z_i$ for all $r = 1, \dots, R$. Again, it is easy to show by induction that $V_{i,r}(x) \geq V_{i,r+1}(x)$, which implies $Z_{i,r}(x) \geq Z_{i,r+1}(x)$. For any x such that $A \cdot \Phi_i(x) - Z_{i,r}(x) \geq 0$, by $Z_{i,r}(x) \geq Z_{i,r+1}(x)$ we have $A \cdot \Phi_i(x) - Z_{i,r+1}(x) \geq A \cdot \Phi_i(x) - Z_{i,r}(x) \geq 0$. Hence, we have $z_{i,r+1} \leq z_{i,r}$, for $r = 1, \dots, R-1$. Take any $\hat{x}_{i,r} \geq z_{i,r+1}$,

$$\begin{aligned} A \cdot \Phi_i(\hat{x}_{i,r}) - Z_{i,r}(\hat{x}_{i,r}) &= A \cdot \Phi_i(\hat{x}_{i,r}) - \mathbb{E}[V_{i,r+1}(\max\{\hat{x}_{i,r}, X_{i,r}\} \mid \tau_{-i})] + c \\ &= A \cdot \Phi_i(\hat{x}_{i,r}) - \mathbb{E}[\max\{A \cdot \Phi_i(\max\{\hat{x}_{i,r}, X_{i,r}\}), Z_{i,r+1}(\max\{\hat{x}_{i,r}, X_{i,r}\} \mid \tau_{-i})\}] + c. \end{aligned}$$

Given that $\max\{\hat{x}_{i,r}, X_{i,r}\} \geq \hat{x}_{i,r} \geq z_{i,r+1}$, we have $A \cdot \Phi_i(\max\{\hat{x}_{i,r}, X_{i,r}\}) \geq Z_{i,r+1}(\max\{\hat{x}_{i,r}, X_{i,r}\})$. Hence,

$$\begin{aligned} A \cdot \Phi_i(\hat{x}_{i,r}) - Z_{i,r}(\hat{x}_{i,r}) &= A \cdot \Phi_i(\hat{x}_{i,r}) - \mathbb{E}[A \cdot \Phi_i(\max\{X_{i,r}, \hat{x}_{i,r}\})] + c \\ &= A \cdot \Phi_i(\hat{x}_{i,r}) - A \cdot \mathbb{E}[\max\{\Phi_i(X_{i,r}), \Phi_i(\hat{x}_{i,r})\}] + c \\ &= -A \cdot \mathbb{E}[(\Phi_i(X_{i,r}) - \Phi_i(\hat{x}_{i,r}))^+] + c. \\ &= A \cdot \Phi_i(\hat{x}_{i,r}) - Z_{i,R}(\hat{x}_{i,r}). \end{aligned}$$

Note that $z_{i,R} \leq z_{i,R+1}$, $A \cdot \Phi_i(z_{i,R}) - Z_{i,R}(z_{i,R}) \geq 0$, and it has been proved that $A \cdot \Phi_i(x) - Z_{i,R}(x)$ is increasing in x in any round $r = 1, \dots, R$. Thus, $A \cdot \Phi_i(z_{i,R+1}) - Z_{i,R}(z_{i,R+1}) = A \cdot \Phi_i(z_{i,R+1}) - Z_{i,R-1}(z_{i,R+1}) \geq 0$. Recall the definition of $z_{i,r}$: $z_{i,r} = \min \{\hat{x}_{i,r} : A \cdot \Phi_i(\hat{x}_{i,r}) - Z_{i,r}(\hat{x}_{i,r}) \geq 0\}$, we must have $z_{i,r+1} \geq z_{i,r}$, for $r = 1, \dots, R-1$. Hence, $z_{i,r} = z_i = z_{i,R} = \min \{\hat{x} \in [0, 1] : A \cdot \mathbb{E}[(\Phi_i(X) - \Phi_i(\hat{x}))^+] \leq c\}$ for all $r = 1, \dots, R$. $\square$

Proof of Theorem 1. We first prove part (i) for $R \rightarrow \infty$. To highlight the dependence on n , let $\Pi_P(n)$ and $\Pi_S(n)$ denote the buyer's expected profit under parallel and sequential search, respectively. Showing $\Pi_S > \Pi_P$ for any $c < A/n$ and $n \in [2, \dots, A/c)$ is equivalent to showing $\Pi_S(n) > \Pi_P(n)$ for any $n \in [2, \dots, A/c)$ and $c > 0$. Define $v = \mathbb{E}[X_{(A/c)}] - A$. The proof has two steps: (a) show $\Pi_S(n) > v$ for all $n \in [2, \dots, A/c)$; (b) show $v \geq \Pi_P(n)$ for all $n \in [2, \dots, A/c)$.

As a preliminary step, we derive the equilibrium stopping threshold z^* with $R \rightarrow \infty$.

LEMMA A5. *When suppliers adopt sequential search, the unique symmetric equilibrium has stopping threshold $z^* = F^{-1}(1 - cn/A)$ as $R \rightarrow \infty$.*

Proof of Lemma A5. As $R \rightarrow \infty$, $G(x; z, \infty) = \lim_{R \rightarrow \infty} F(x)^R = 0$ for $x \leq z$, and $G(x; z, \infty) = [F(x) - F(z)]/[1 - F(z)]$ for $x > z$. Hence (5) simplifies to

$$A \int_z^1 \left[\left(\frac{F(x) - F(z)}{1 - F(z)} \right)^{n-1} \right] dF(x) - c = 0 \iff \frac{A}{n[1 - F(z)]^{n-1}} [F(x) - F(z)]^n \Big|_z^1 - c = 0$$

which yields $A/n \cdot [1 - F(z)] - c = 0$, i.e., $F(z) = 1 - \frac{cn}{A}$. Thus $z^* = F^{-1}(1 - cn/A)$ as $R \rightarrow \infty$. $\square$

By Lemma A5, when $R \rightarrow \infty$ the CDF of a single supplier's best performance is

$$G(x; z^*, \infty) = \begin{cases} 0, & x \leq z^*, \\ \frac{F(x) - 1 + cn/A}{cn/A}, & x > z^*. \end{cases}$$

Let $L(x; n)$ be the CDF of $\max_{i=1, \dots, n} X_{i,(\tau_i^*)}$; then $L(x; n) = G(x; z^*, \infty)^n$, i.e.,

$$L(x; n) = \begin{cases} 0, & x \leq z^*, \\ \left(\frac{F(x) - 1 + cn/A}{cn/A} \right)^n, & x > z^*. \end{cases}$$

(a): $\Pi_S(n) > v$. It suffices to show $\max_i X_{i,(\tau_i^*)}$ first-order stochastically dominates $X_{(A/c)}$, i.e., $L(x; n) \leq F(x)^{A/c}$ with strict inequality on a set of positive measure. For $x \leq z^*$ the strict inequality trivially holds. For $x > z^*$, it is equivalent to show

$$\left(\frac{F(x) - 1 + cn/A}{cn/A} \right)^n \leq F(x)^{A/c} \iff \frac{A}{cn} \left[F(x) - 1 + \frac{cn}{A} - \frac{cn}{A} F(x)^{A/(cn)} \right] \leq 0.$$

The term $F(x) - 1 + (cn)/A - (cn)/A \cdot F(x)^{A/(cn)}$ is increasing in $F(x) \in [0, 1]$, and it equals to 0 when $F(x) = 1$. Hence the inequality holds for all $x \in [z^*, 1)$.

(b): $v \geq \Pi_P(n)$. As a preliminary step, we first argue that under parallel search the total expected number of searches $\bar{m} = \mathbb{E}[\sum_{i=1}^n m_i^*] = n \cdot \mathbb{E}[m^*] < A/c$. By the equilibrium structure characterized in Proposition 1 (i) and (ii), we have

$$\mathbb{E}[m^*] \leq m_0 + \frac{1}{n} = \frac{A(n-1)}{cn^2} + \frac{1}{n} = \frac{A}{cn} \left(\frac{n-1}{n} + c/A \right) < \frac{A}{cn} \left(\frac{n-1}{n} + \frac{1}{n} \right) = \frac{A}{cn}.$$

Hence, $\bar{m} = n \cdot \mathbb{E}[m^*] < A/c$.

Therefore

$$\Pi_P(n) = \mathbb{E} \left[\max_i X_{i, (m_i^*)} \right] - A = \mathbb{E} \left[\mathbb{E} \left[X_{(\sum_i m_i^*)} \mid \sum_i m_i^* \right] \right] - A \stackrel{(a)}{\leq} \mathbb{E} [X_{(\mathbb{E}[\sum_i m_i^*])}] - A \stackrel{(b)}{\leq} \mathbb{E} [X_{(A/c)}] - A = v,$$

in which (a) is by Jensen's Inequality and the fact that $\mathbb{E}[X_{(m)}]$ is concave in m and (b) is due to $\mathbb{E}[X_{(m)}]$ is increasing in m . This completes our proof for part (i).

For part (ii), fix $R < \infty$ and we use the notation $\Pi_P(c)$ and $\Pi_S(c)$ to emphasize the dependence on c . Under parallel search, as $c \rightarrow 0$, $m_0 \rightarrow \infty$ and m_0 decreases in c , so there exists $c_l > 0$ such that $R < \lfloor m_0 \rfloor$ for all $c \in (0, c_l)$; by Proposition 1(iii), $m^* = R$ for $c \in (0, c_l)$. As $c \rightarrow A/n$, $[m_0 - 1/n, m_0 + 1/n] \rightarrow [(n-2)/n, 1]$; since m_0 decreases in c , there exists $c_h > 0$ with $1 \in [m_0 - 1/n, m_0 + 1/n]$ for all $c \in (c_h, A/n)$, and by Proposition 1(i), $m^* = 1$ for such c .

Next, for fixed R , we prove $\Pi_S(c)$ is strictly decreasing in c . It suffices to show that $G(x; z^*, R)$ increases in c by: (a) z^* decreases strictly in c and (b) $F(z^*)^R + [1 - F(z^*)^R] \cdot \frac{F(x) - F(z^*)}{1 - F(z^*)}$ decreases strictly in z^* .

For (a), define

$$h(z) = A \int_z^1 (G(x; z, R)^{n-1} - G(z; z, R)^{n-1}) dF(x) - c.$$

Differentiating yields

$$h'(z) = A \int_z^1 \left\{ (n-1)G(x; z, R)^{n-2} \frac{1 - F(x)}{[1 - F(z)]^2} l(R, z) f(z) - (n-1)G(z; z, R)^{n-2} R F(z)^{R-1} f(z) \right\} dF(x),$$

where $l(R, z) = R F(z)^{R-1} - (R-1)F(z)^R - 1 \leq 0$ (since $\partial l / \partial z \geq 0$ and $l(R, 1) = 0$). Hence $h'(z) < 0$, so z^* decreases in c .

For (b),

$$F(z^*)^R + [1 - F(z^*)^R] \frac{F(x) - F(z^*)}{1 - F(z^*)} = 1 - \sum_{i=0}^{R-1} F(z^*)^i [1 - F(x)],$$

which is strictly decreasing in $F(z^*)$ (thus in z^*), so it increases in c . Therefore $G(x; z^*, R)$ increases in c for all x , strictly for $x \in (z^*, 1]$.

Thus $X_{i, (\tau_i^*)}$ is strictly decreasing in c in the sense of first-order stochastic dominance for each i , implying

$$\Pi_S(c) = \mathbb{E} \left[\max_i X_{i, (\tau_i^*)} \right] - A$$

is strictly decreasing in c . As $c \rightarrow 0$, $F(z^*) \rightarrow 1$ and suppliers search until round R , so

$$\lim_{c \rightarrow 0} \Pi_S(c) = \mathbb{E} \left[\max_i X_{i, (R)} \right] - A = \lim_{c \rightarrow 0} \Pi_P(c).$$

As $c \rightarrow A/n$, $F(z^*) \rightarrow 0$ and suppliers stop after the first round, so

$$\lim_{c \rightarrow A/n} \Pi_S(c) = \mathbb{E} \left[\max_i X_{i,(1)} \right] - A = \lim_{c \rightarrow A/n} \Pi_P(c).$$

Therefore, for any given R and n , $\Pi_S(c) \rightarrow \Pi_P(c)$ as $c \rightarrow 0$ and as $c \rightarrow A/n$, and since $\Pi_S(c)$ is strictly decreasing in c , $\Pi_S > \Pi_P$ for $c \in (c_h, A/n)$ and $\Pi_S < \Pi_P$ for $c \in (0, c_l)$. $\square$

Proof of Lemma 2. In order to prove $\Pi_S \geq \Pi_0$, it suffices to show that $\max_{i=1,\dots,n} X_{i,(\tau_i^*)}$ dominates $\max_{i=1,\dots,n} X_{i,(\tilde{m})}$ in the first-order stochastic dominance sense, where $\tilde{m}$ follows a (truncated) geometric distribution with $\mathbb{P}(\tilde{m} = k) = F(z^*)^{k-1} [1 - F(z^*)]$ for $k \in \{1, \dots, R-1\}$ and $\mathbb{P}(\tilde{m} = R) = F(z^*)^{R-1}$. Since first-order stochastic dominance is preserved when taking maximum over n i.i.d. random variables, it is therefore sufficient to prove that $X_{i,(\tau_i^*)}$ dominates $X_{i,(\tilde{m})}$ in the first-order stochastic dominance sense.

From (4), the cumulative distribution function of $X_{i,(\tau_i^*)}$ is given by

$$G(x; z^*, R) = \begin{cases} F(x)^R, & x \leq z^*, \\ F(z^*)^R + [1 - F(z^*)^R] \frac{F(x) - F(z^*)}{1 - F(z^*)}, & x > z^*. \end{cases}$$

We denote by $H(x; z^*, R)$ the cumulative distribution function of $\max_{i=1,\dots,n} X_{i,(\tilde{m})}$. It follows from the distribution of $\tilde{m}$ that

$$H(x; z^*, R) = \sum_{k=1}^{R-1} F(z^*)^{k-1} [1 - F(z^*)] F(x)^k + F(z^*)^{R-1} F(x)^R.$$

To establish first-order stochastic dominance, it suffices to show that

$$H(x; z^*, R) - G(x; z^*, R) \geq 0 \quad \text{for all } x \in [0, 1].$$

First, consider case (a) in which $x \leq z^*$. We have

$$\begin{aligned} H(x; z^*, R) - G(x; z^*, R) &= \sum_{k=1}^{R-1} F(z^*)^{k-1} [1 - F(z^*)] F(x)^k + F(z^*)^{R-1} F(x)^R - F(x)^R \\ &= \sum_{k=1}^{R-1} F(z^*)^{k-1} [1 - F(z^*)] F(x)^k + F(x)^R [F(z^*)^{R-1} - 1] \\ &= \sum_{k=1}^{R-1} F(z^*)^{k-1} [1 - F(z^*)] F(x)^k + F(x)^R \left\{ \sum_{k=1}^{R-1} F(z^*)^{k-1} [1 - F(z^*)] \right\} \\ &= \sum_{k=1}^{R-1} F(z^*)^{k-1} [1 - F(z^*)] [F(x)^k - F(x)^R] \geq 0, \end{aligned}$$

since $F(x)^k \geq F(x)^R$ for all $k \leq R-1$.

Next, consider case (b), where $x > z^*$. In this case, we have

$$\begin{aligned} H(x; z^*, R) - G(x; z^*, R) &= \sum_{k=1}^{R-1} F(z^*)^{k-1} [1 - F(z^*)] F(x)^k + F(z^*)^{R-1} F(x)^R \\ &\quad - F(z^*)^R - [1 - F(z^*)^R] \frac{F(x) - F(z^*)}{1 - F(z^*)}. \end{aligned}$$

If $R = 1$, then

$$H(x; z^*, 1) - G(x; z^*, 1) = F(x) - F(z^*) - F(x) + F(z^*) = 0.$$

Next, we show that

$$[H(x; z^*, R+1) - G(x; z^*, R+1)] - [H(x; z^*, R) - G(x; z^*, R)] \geq 0, \quad \forall R \geq 1,$$

which implies that $H(x; z^*, R) - G(x; z^*, R) \geq 0$ for all $R \geq 1$ and $x > z^*$. Indeed,

$$\begin{aligned} & [H(x; z^*, R+1) - G(x; z^*, R+1)] - [H(x; z^*, R) - G(x; z^*, R)] \\ &= F(z^*)^{R-1} [1 - F(z^*)] F(x)^R + F(z^*)^R F(x)^{R+1} - F(z^*)^{R-1} F(x)^R - F(z^*)^{R+1} + F(z^*)^R \\ & - [1 - F(z^*)^{R+1}] \frac{F(x) - F(z^*)}{1 - F(z^*)} + [1 - F(z^*)^R] \frac{F(x) - F(z^*)}{1 - F(z^*)} \\ &= F(z^*)^R [F(x)^{R+1} - F(x)^R - F(z^*) + 1 - F(x) + F(z^*)] \\ &= F(z^*)^R [1 - F(x)] [1 - F(x)^R] \geq 0. \end{aligned}$$

Hence,

$$H(x; z^*, R) - G(x; z^*, R) \geq 0 \quad \text{for all } R \geq 1 \text{ and } x > z^*.$$

Combining cases (a) and (b) completes the proof. $\square$

Proof of Proposition 3. We begin with part (i) and first establish the claim for $\Pi_P(R)$. It suffices to show that the equilibrium $m^*(R)$ is increasing in R in the sense of first-order stochastic dominance; that is, for any $R_1 < R_2$, $m^*(R_2)$ first-order stochastically dominates $m^*(R_1)$. If $R_1 \geq \lceil m_0 \rceil$, then $R_2 \geq \lceil m_0 \rceil$ and hence $m^*(R_1) = m^*(R_2)$. If $R_1 < \lceil m_0 \rceil$ and $R_2 < \lceil m_0 \rceil$, then $m^*(R_1) = R_1 < R_2 = m^*(R_2)$. Finally, if $R_1 < \lceil m_0 \rceil \leq R_2$, then $m^*(R_1) = R_1$, while $m^*(R_2)$ equals either $\lfloor m_0 \rfloor$ or $\lceil m_0 \rceil$, or is a random variable that takes the value $\lfloor m_0 \rfloor$ with probability p^* and $\lceil m_0 \rceil$ with probability $1 - p^*$. Since $\lceil m_0 \rceil > \lfloor m_0 \rfloor \geq R_1$, we also have $m^*(R_2)$ first-order stochastically dominates $m^*(R_1)$. Thus $\Pi_P(R)$ is increasing in R .

We next establish the claim for $\Pi_S(R)$ by showing that $\Pi_S(R) \leq \Pi_S(R+1)$. Let z_R^* denote the solution to equation (5) in Proposition 2, and let $\tau_i^*(R)$ denote the corresponding equilibrium stopping time when the number of evaluation opportunities is R . We note that, by direct calculation, z_R^* equivalently solves

$$[1 - F(z)] \left[\frac{1 - F(z)^{Rn}}{1 - F(z)^R} - nF(z)^{R(n-1)} \right] = \frac{cn}{A}. \quad (13)$$

It suffices to show that $X_{i, \tau_i^*(R+1)}$ first-order stochastically dominates $X_{i, \tau_i^*(R)}$, since this implies that $\max_i X_{i, \tau_i^*(R+1)}$ also first-order stochastically dominates $\max_i X_{i, \tau_i^*(R)}$, and hence $\Pi_S(R+1) \geq \Pi_S(R)$.

Recall that the CDF of $X_{i, \tau_i^*(R)}$ is

$$G(x; z_R^*, R) = \begin{cases} F(x)^R, & x \leq z_R^*, \\ F(z_R^*)^R + [1 - F(z_R^*)^R] \frac{F(x) - F(z_R^*)}{1 - F(z_R^*)}, & x > z_R^*, \end{cases}$$

The rest of the proof is based on the following lemma.

LEMMA A6. Let $n \geq 2$, $R \geq 1$, and $\frac{cn}{A} \in (0, 1)$. Let $x^*(R)$ be the unique solution to the equation

$$(1-x) \left(\frac{1-x^{Rn}}{1-x^R} - n x^{R(n-1)} \right) = \frac{cn}{A}. \quad (14)$$

then

$$\frac{1-x^*(R)^R}{1-x^*(R)} < \frac{1-x^*(R+1)^{R+1}}{1-x^*(R+1)}.$$

Proof of Lemma A6. Define

$$H(u) := (1-u) \left(\frac{1-u^n}{1-u} - n u^{n-1} \right) = 1 - n u^{n-1} + (n-1) u^n,$$

which is strictly decreasing on $(0, 1)$. For $R \geq 1$, let

$$I_R(u) := \frac{1-u}{1-u^{1/R}} = \sum_{k=0}^{R-1} u^{k/R},$$

which is strictly increasing in u and satisfies $I_{R+1}(u) > I_R(u)$ for all $u \in (0, 1)$. Let $w := x^*(R)^R$ and $y := x^*(R+1)^{R+1}$. By (14),

$$\frac{H(w)}{I_R(w)} = \frac{cn}{A} \quad \text{and} \quad \frac{H(y)}{I_{R+1}(y)} = \frac{cn}{A}.$$

If $w \leq y$, then $H(w) \geq H(y)$, and moreover

$$I_{R+1}(y) \geq I_{R+1}(w) > I_R(w),$$

which implies

$$\frac{H(w)}{I_R(w)} > \frac{H(y)}{I_{R+1}(y)},$$

a contradiction. Hence $w > y$, which implies $H(y) > H(w)$ and therefore

$$I_{R+1}(y) = \frac{H(y)}{cn/A} > \frac{H(w)}{cn/A} = I_R(w),$$

that is,

$$\frac{1-x^*(R+1)^{R+1}}{1-x^*(R+1)} > \frac{1-x^*(R)^R}{1-x^*(R)}.$$

□

By Lemma A6 and (13),

$$\frac{1-F(z_R^*)^R}{1-F(z_R^*)} < \frac{1-F(z_{R+1}^*)^{R+1}}{1-F(z_{R+1}^*)}. \quad (15)$$

Next, we distinguish between (a) $z_R^* \leq z_{R+1}^*$; and (b) $z_R^* > z_{R+1}^*$, and show that in either case,, and show that in either case, $G(x; z_{R+1}^*, R+1) \leq G(x; z_R^*, R)$ for all $x \in [0, 1]$, with strict inequality holding on a subset of $[0, 1]$ of positive measure.

Consider case (a) when $z_R^* \leq z_{R+1}^*$. If $x \leq z_R^* \leq z_{R+1}^*$, we clearly have $G(x; z_{R+1}^*, R+1) = F(x)^{R+1} < F(x)^R = G(x; z_R^*, R)$. If $x \in (z_R^*, z_{R+1}^*]$, we aim to prove the following:

$$\begin{aligned}
F(x)^{R+1} &\leq F(z_R^*)^R + [1 - F(z_R^*)^R] \frac{F(x) - F(z_R^*)}{1 - F(z_R^*)} \\
&\Leftrightarrow F(x)^{R+1} \leq 1 - \frac{1 - F(z_R^*)^R}{1 - F(z_R^*)} \cdot [1 - F(x)] \\
&\Leftrightarrow \frac{1 - F(z_R^*)^R}{1 - F(z_R^*)} \leq \frac{1 - F(x)^{R+1}}{1 - F(x)}.
\end{aligned}$$

First note that $(1 - F(x)^{R+1})/(1 - F(x)) \geq (1 - F(x)^R)/(1 - F(x))$ and as $(1 - F(x)^R)/(1 - F(x))$ is increasing in x , $(1 - F(x)^R)/(1 - F(x)) \geq (1 - F(z_R^*)^R)/(1 - F(z_R^*))$. Hence, $(1 - F(x)^{R+1})/(1 - F(x)) \geq (1 - F(z_R^*)^R)/(1 - F(z_R^*))$, for any $x \in (z_R^*, z_{R+1}^*]$.

If $x \in (z_{R+1}^*, 1]$, we aim to prove the following:

$$\begin{aligned}
&F(z_{R+1}^*)^{R+1} + [1 - F(z_{R+1}^*)^{R+1}] \frac{F(x) - F(z_{R+1}^*)}{1 - F(z_{R+1}^*)} \leq F(z_R^*)^R + [1 - F(z_R^*)^R] \frac{F(x) - F(z_R^*)}{1 - F(z_R^*)} \\
&\Leftrightarrow 1 - \frac{1 - F(z_{R+1}^*)^{R+1}}{1 - F(z_{R+1}^*)} \cdot [1 - F(x)] \leq 1 - \frac{1 - F(z_R^*)^R}{1 - F(z_R^*)} \cdot [1 - F(x)] \\
&\Leftrightarrow \frac{1 - F(z_R^*)^R}{1 - F(z_R^*)} \leq \frac{1 - F(z_{R+1}^*)^{R+1}}{1 - F(z_{R+1}^*)},
\end{aligned} \tag{16}$$

which is true by (15).

Next consider case (b) when $z_R^* > z_{R+1}^*$. If $x \leq z_{R+1}^* < z_R^*$, we clearly have $G(x; z_{R+1}^*, R+1) = F(x)^{R+1} < F(x)^R = G(x; z_R^*, R)$. If $x \in (z_{R+1}^*, z_R^*]$, we aim to prove the following:

$$\begin{aligned}
&F(z_{R+1}^*)^{R+1} + [1 - F(z_{R+1}^*)^{R+1}] \frac{F(x) - F(z_{R+1}^*)}{1 - F(z_{R+1}^*)} \leq F(x)^R \\
&\Leftrightarrow 1 - \frac{1 - F(z_{R+1}^*)^{R+1}}{1 - F(z_{R+1}^*)} \cdot [1 - F(x)] \leq F(x)^R \\
&\Leftrightarrow \frac{1 - F(x)^R}{1 - F(x)} \leq \frac{1 - F(z_{R+1}^*)^{R+1}}{1 - F(z_{R+1}^*)}.
\end{aligned}$$

Note that $(1 - F(x)^R)/(1 - F(x)) \leq (1 - F(z_R^*)^R)/(1 - F(z_R^*))$. By (15), we thus have $(1 - F(x)^R)/(1 - F(x)) \leq (1 - F(z_R^*)^R)/(1 - F(z_R^*)) < (1 - F(z_{R+1}^*)^{R+1})/(1 - F(z_{R+1}^*))$, for any $x \in (z_{R+1}^*, z_R^*]$. If $x \in (z_R^*, 1]$, the argument follows the same as (16).

Next we move on to part (ii). We use the notation $\Pi_S(n)$ to emphasize the dependence of the buyer's expected profit on n when suppliers adopt sequential search strategies. Recall from our proof of Proposition 1 that the CDF of the best performance value submitted by n suppliers as $R \rightarrow \infty$

$$L(x; n) = \begin{cases} 0, & x \leq z^* \\ \left(\frac{F(x) - 1 + cn/A}{cn/A} \right)^n, & x > z^*. \end{cases}$$

We first prove $L(x; n)$ is increasing in n for $x > z^*$, which is equivalent as showing $\ln(L(x; n))$ is increasing in n for $x > z^*$. Note that $\ln(L(x; n))$ is increasing in n in the integer domain if its natural continuous extension

is also increasing. Hence, in the following we treat n as a continuous variable. Taking first order derivative of $\ln(L(x; n))$ with respect to n , we have for $x > z^*$:

$$\begin{aligned} \frac{\partial \ln(L(x; n))}{\partial n} &= \frac{\partial \left(n \cdot \ln \left[\frac{F(x) - 1 + cn/A}{cn/A} \right] \right)}{\partial n} \\ &= \ln \left[\frac{F(x) - 1 + cn/A}{cn/A} \right] + n \cdot \frac{cn/A}{F(x) - 1 + cn/A} \cdot \frac{c/A \cdot cn/A - c/A \cdot [F(x) - 1 + cn/A]}{(cn/A)^2} \\ &= \ln \left[\frac{F(x) - 1 + cn/A}{cn/A} \right] + \frac{1 - F(x)}{F(x) - 1 + cn/A} \\ &= \ln \left[\frac{F(x) - 1 + cn/A}{cn/A} \right] + \frac{cn/A}{F(x) - 1 + cn/A} - 1. \end{aligned}$$

Note that $F(z^*) = 1 - cn/A$, and with any $F(z^*) < F(x)$, $(F(x) - 1 + cn/A)/(cn/A) \in (0, 1]$. In addition, the function $g(y) = \ln y + 1/y - 1$ is decreasing in y since $g'(y) = 1/y - 1/(y^2) \leq 0$ on $(0, 1]$ with $g(1) = 0$. As a result, we must have $\partial \ln(L(x; n))/\partial n \geq 0$ for $x > z^*$.

Given that $L(x; n)$ is increasing in n for $x > z^*$ and the fact that z^* is decreasing in n , we then have $L(x; n)$ is increasing in n for all x . That is, the random variable $\max_{i=1, \dots, n} X_{i, (\tau_i^*)}$ is decreasing in n in the sense of first-order stochastic dominance. It then follows that $\mathbb{E} \left[\max_{i=1, \dots, n} X_{i, (\tau_i^*)} \right]$ is decreasing in n and hence $\Pi_S(n)$ is decreasing in n . $\square$

Proof of Theorem 2. By Proposition 3(i), we always have $R^* \rightarrow \infty$ regardless of the number of suppliers and whether they engage in sequential search or parallel search. Given $R^* \rightarrow \infty$, by Theorem 1(i), it is always optimal to require all suppliers to engage in sequential search regardless of the number of suppliers. Finally, Proposition 3(ii) implies that the optimal number of suppliers is $n^* = 2$. $\square$

Proof of Theorem 3. First note that $\Pi_S^e - \Pi_P^e = (\Pi_S - \Pi_P) - c_e(n\mathbb{E}[\tau_i^*] - \mathbb{E}[nm^*])$ is linear in c_e , its sign changes at most once as c_e increases. We first consider part (i), where $R \rightarrow \infty$. By Theorem 1(i), $\Pi_S > \Pi_P$ for all $c \in (0, A/n)$. By Lemma A5, $n\mathbb{E}[\tau_i^*] = A/c$, and $\mathbb{E}[nm^*] < A/c$ as shown in the proof of Theorem 1. Hence, $\Pi_S^e - \Pi_P^e$ is strictly decreasing in c_e , so there exists a unique $\bar{c}_e^\infty(c) > 0$ such that $\Pi_S^e < \Pi_P^e$ if $c_e > \bar{c}_e^\infty(c)$.

We next consider part (ii). Where $R < \infty$ and $c \in (c_h, A/n)$, by Theorem 1(ii), $\Pi_S > \Pi_P$. Proposition 1(i) gives $m^* = 1$, so $\mathbb{E}[nm^*] = n$. Since $F(z^*) > 0$ for any $c \in (0, A/n)$, so $\mathbb{E}[\tau_i^*] = \sum_{k=0}^{R-1} F(z^*)^k > 1$, giving $n\mathbb{E}[\tau_i^*] - \mathbb{E}[nm^*] > 0$. Hence $\Pi_S^e - \Pi_P^e$ is strictly decreasing in c_e . So there exists a unique $\bar{c}_e^R(c) > 0$ such that $\Pi_S^e < \Pi_P^e$ if $c_e > \bar{c}_e^R(c)$. Finally, when $R < \infty$ and $c \in (0, c_l)$, by Theorem 1(ii), $\Pi_P > \Pi_S$. Proposition 1(iii) gives $m^* = R$, so $\mathbb{E}[nm^*] = nR$. Since $F(z^*) < 1$, so $\mathbb{E}[\tau_i^*] = (1 - F(z^*)^R)/(1 - F(z^*)) < R$, giving $n\mathbb{E}[\tau_i^*] - \mathbb{E}[nm^*] < 0$. Hence $\Pi_S^e - \Pi_P^e$ is strictly increasing in c_e . So there exists a unique $\bar{c}_e^R(c) > 0$ such that and $\Pi_P^e < \Pi_S^e$ if $c_e > \bar{c}_e^R(c)$. $\square$